

Cognitive Frictions in Decisions under Risk: Evidence from (Almost) Degenerate Lottery Evaluation

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Abstract

We examine cognitive frictions in decisions under risk using a test based on (almost) degenerate lottery evaluation. While standard theories predict their valuations converge to degenerate payoffs, we find that most participants fail this test in price-list experiments. Test failure correlates with a more pronounced fourfold pattern of risk attitudes, robust to standard cognitive controls. This association attenuates among university students relative to Prolific samples and vanishes under binary choices versus price lists. Our findings indicate that canonical anomalies attributed to lottery-specific evaluation may partly reflect general cognitive constraints, highlighting that cognitive frictions operate through multifaceted channels to shape decision-making.

Keywords: risk attitudes, fourfold pattern, preference, non-expected utility, cognitive frictions, complexity, experiment

JEL classification: C81, C91

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1 Introduction

How individuals evaluate risky prospects lies at the core of the study of decisions under risk. A well-documented empirical regularity is the fourfold pattern of risk attitudes: risk seeking for low-probability gains and high-probability losses, and risk aversion for high-probability gains and low-probability losses (Kahneman and Tversky, 1979). This pattern has become a cornerstone in behavioral economics and has motivated various theories and interpretations that can be categorized into two approaches. The first, classical approach generalizes expected utility (EU) and posits non-standard *preferences over lotteries*.¹ The models following this approach typically feature an inverse S-shaped probability weighting function that overweighs small probabilities and underweighs moderate to large probabilities, which accounts for a range of behavioral anomalies including the fourfold pattern. A second, recent approach focuses on various aspects of *cognition about lotteries* when integrating probabilities with payoffs.² For example, individuals may struggle with contingent reasoning, allocate attention disproportionately to salient payoffs and probabilities, miscalculate expected payoffs, and use imprecise mental representations. As a result of these cognitive frictions, individuals often rely on heuristic rules or make stochastic choices, giving rise to a wide range of behavioral anomalies. Both approaches and their related experimental tests emphasize lottery-specific characteristics.

Here we examine cognitive friction in decisions under risk using a simple test based on (almost) degenerate lottery evaluation (hereafter, the ADLE test) that minimizes the impact of lottery-specific characteristics. We assess its performance in relation to the fourfold pattern of risk attitudes. Experimental studies of the fourfold pattern usually involve the measurement of the certainty equivalents (CEs) of binary lotteries of the form $(x, p; 0)$ that deliver outcome x with probability p and 0 otherwise. In the ADLE test, we

¹See, for example, Kahneman and Tversky (1979); Quiggin (1982); Chew (1983); Gul (1991); Cerreia-Vioglio et al. (2015); Masatlioglu and Raymond (2016). See Starmer (2000); Wakker (2010) for reviews.

²See, for example, Shafir and Tversky (1992); Bordalo et al. (2012); Martínez-Marquina et al. (2019); Khaw et al. (2021); Enke and Graeber (2023); Esponda and Vespa (2024); Oprea (2024); Puri (2025); Enke et al. (2026). See reviews on related topics: Gabaix (2019); Maćkowiak et al. (2023); Loewenstein and Wojtowicz (2025) on attention/inattention, Woodford (2020) on imprecision, Bordalo et al. (2012) on salience, Niederle and Vespa (2023) on contingent thinking, de Clippel and Rozen (2024) on bounded rationality, Enke (2024) on cognition, and Oprea (2025) on complexity.

replace commonly used lotteries with test lotteries, namely, degenerate lotteries $(x, p; x)$ and almost-degenerate lotteries $(x + \epsilon, p; x - \epsilon)$ for small ϵ . Both aforementioned approaches would imply that individuals evaluate (almost) degenerate lotteries close to their degenerate values.

We examine the relationship between ADLE test performance and the fourfold pattern in a series of experiments. In the main experiment, we elicit CEs of ten standard lotteries and eight test lotteries using price lists from a Prolific sample. All lotteries are represented by 100 boxes, with participants paid the amount in a randomly selected box. The 10 standard lotteries have $100p$ boxes containing $\pm \$25$ and $100(1 - p)$ containing $\$0$, $p = 0.1, 0.25, 0.5, 0.75, 0.9$, where $\pm$ denotes gain and loss frames. The eight test lotteries are represented using the same 100-box structure: degenerate lotteries $\pm(\$2.30, 0.10; \$2.30)$ and $\pm(\$22.70, 0.90; \$22.70)$, and almost degenerate lotteries $\pm(\$2.90, 0.10; \$2.10)$ and $\pm(\$22.90, 0.90; \$22.10)$. We refer to the valuation of a single test lottery as a test task; a participant fails the ADLE test if the CE differs from the correct value in at least one test task.

Our analysis yields three main findings. First, the majority of participants fail the ADLE test, and most fail multiple test tasks. Second, participants who fail the test display a more pronounced fourfold pattern than those who pass. Third, performance on comprehension questions neither strongly predicts test failure nor explains its association with the fourfold pattern.

Given the widespread failure of the ADLE test and its association with the fourfold pattern, we implement two additional experiments to incorporate two recently developed measures of *cognition about lottery* that have been shown to be linked to the fourfold pattern. The first additional experiment incorporates the valuation of *deterministic mirrors*, as proposed by Oprea (2024). Deterministic mirrors share the same structure as the lotteries, but their payoff equals the average amount of all 100 boxes rather than the amount in a randomly chosen box. Deterministic mirrors have objectively defined values like our test lotteries, but require an extra step of computing the average. In addition to replicating the main observations in Oprea (2024), we find that, in both lotteries and deterministic mirrors, the fourfold pattern is stronger among participants who fail the ADLE test than

among those who pass. Importantly, when both are jointly considered, the fourfold pattern is more strongly linked to ADLE test failure than to the valuation of deterministic mirrors, suggesting that the extra step of computing the average plays only a limited additional role in driving the fourfold pattern.

The second additional experiment incorporates the elicitation of *cognitive uncertainty* (Enke and Graeber, 2023), in which participants report how certain they are about their CEs of the lotteries. We find that the link between the fourfold pattern and the failure of the ADLE test holds in both the high- and low-cognitive-uncertainty subsamples and is robust to controlling for cognitive uncertainty. Furthermore, when we examine cognitive uncertainty for the degenerate lotteries and the standard lotteries simultaneously, uncertainty about the former is more strongly associated with the fourfold pattern. This observation suggests that, in our simple setting with binary lotteries, lottery-specific cognitive uncertainty plays a limited role, aligning with the findings from the deterministic-mirror experiment.

To gain further insight into the relationship between ADLE test performance and the fourfold pattern, we conduct a third additional experiment replacing the Prolific sample with a student sample from a Dutch research university. The use of a student sample allows us to evaluate the generalizability of our findings. In contrast to the Prolific sample, most student participants pass the ADLE test, and those who fail typically fail only one test task, suggesting that student participants may be cognitively more attentive in the experimental environment. Nevertheless, as in the Prolific sample, the fourfold pattern is stronger among those who fail than among those who pass, and this difference remains robust when we control for performance on comprehension questions.

Finally, we conduct the fourth additional experiment replacing the price lists with standalone binary choices in a Prolific sample. For each standard and test lottery, we extract from its corresponding price list two binary choices between the lottery and the two sure amounts nearest its expected value. Because binary choices are arguably simpler than price lists, we expect both ADLE test failures and their association with the fourfold pattern to attenuate. Consistent with this hypothesis, participants are substantially less likely to fail the ADLE test and to exhibit the fourfold pattern than in the main experiment —

echoing the findings in Harbaugh et al. (2010) and Bouchouicha et al. (2026). Moreover, the association between ADLE test failure and the fourfold pattern no longer holds.

Taken together, the first two additional experiments reinforce the findings from our main study. A substantial share of participants fail the ADLE test, and failure is systematically associated with a more pronounced fourfold pattern of risk preferences — an effect that remains robust after controlling for performance on comprehension questions, behavior on deterministic mirrors, and cognitive uncertainty. The remaining two experiments delineate two boundary conditions for these findings: both the failure rate and the magnitude of the association are attenuated when using a university student sample rather than a Prolific sample, and when replacing the price lists with binary choices.

These observations add to the literature on decisions under risk. First, the preference-over-lotteries approach is central to the modeling of decisions under risk. By incorporating an inverse S-shaped probability weighting function, most such models generalize expected utility to rationalize canonical behavioral anomalies. While these frameworks do not explicitly formalize cognitive processes, they draw on core psychological intuitions such as diminishing sensitivity and optimism/pessimism (Wakker, 2025). Crucially, however, nearly all such models predict convergence in the ADLE test: the valuation of an (almost) degenerate lottery should approach that of the corresponding expected value. As a result, they are also silent about the link between the ADLE test failure and the fourfold pattern. Rather than serving as an indictment of preference-based models, our study suggests that choice data from participants who persistently fail the test are better classified as noise than treated as valid evidence testing these theories.

Second, our results provide new insights into the cognition-based approach, which has become increasingly prominent in the recent literature. While studies along this direction have primarily emphasized lottery-specific cognitive frictions, such as difficulties in perceiving probabilities or computing expected values (see e.g., Khaw et al., 2021; Enke and Graeber, 2023; Oprea, 2024), our findings point to a broader class of cognitive limitations. In particular, in our deterministic-mirror experiment, we find that failure of the ADLE test is a key driver of the fourfold pattern in both lotteries and mirrors, even after accounting for the computational complexity of lotteries. Consistent with this finding, cognitive un-

certainty related to ADLE is strongly associated with the fourfold pattern after controlling for lottery-specific cognitive uncertainty. Because neither ADLE failure nor ADLE-related cognitive uncertainty involves lottery-specific perception and computation, these results point to the importance of more general, non-lottery-specific cognitive frictions in driving some of the observed behavioral anomalies under risk. In the binary-choice experiment, we document both a sharply lower ADLE test failure rate and a weaker association with the fourfold pattern. This contrast underscores the importance of general cognition about the elicitation mechanism for understanding decision-making under risk.

It is important to note that the relative importance of general versus lottery-specific cognitive frictions likely depends on the complexity of the decision environment. In our setting with simple binary lotteries $(x, p; 0)$, lottery-specific computational demands are limited. In more complex environments, however, such frictions may become more consequential, for instance, when payoffs are state-contingent and must be calculated in bidding tasks (Martínez-Marquina et al., 2019), when probabilities and payoffs are non-round numbers (Khaw et al., 2023), or when lotteries involve large numbers of outcomes (Puri, 2025). Moreover, the two types of friction may reinforce each other. Lottery-specific computational demands can deplete cognitive bandwidth, worsening environment-level frictions. Conversely, environment-level frictions, such as complex elicitation mechanisms, can sharply reduce an individual’s tolerance for even modest lottery-specific computational complexity.

Finally, our study contributes a practical diagnostic tool for researchers designing experiments on risk attitudes. Experimental studies typically include detailed instructions, comprehension tests, incentivized choices, attention checks, and cognitive measures. These design features help enhance motivation, reduce misconceptions about the experiments, and improve data quality (Plott and Zeiler, 2005, 2007; Cason and Plott, 2014). While valuable, these instruments are imperfect proxies for the cognition required to evaluate the lotteries in the experiments. For instance, in our study, the correlation between performance on the ADLE test and the comprehension questions is only moderate. Because the ADLE test is embedded in the same decision environment as the experimental task, it provides a more direct and stringent assessment of whether participants understand the

setting in which they make decisions. The ADLE test is closely related to the mirror test in Oprea (2024) but avoids its computational complexity. Combined with the measure of cognitive uncertainty, the ADLE test also allows us to separately assess cognitive uncertainty regarding (almost) degenerate lotteries and standard lotteries, thereby disentangling environment-level factors from lottery-specific uncertainty. For experimental studies, the ADLE test thus offers a practical, task-embedded diagnostic for assessing the extent to which observed behavior reflects genuine lottery valuation rather than broader features of the decision environment.

The rest of the paper is organized as follows. Section 2 presents the experimental design. Section 3 reports results from the main experiment and the four additional experiments. Section 4 concludes and discusses the broader implications of our findings.

2 Experimental design

This section presents the experimental design. We first outline the tasks in the main experiment, then highlight the differences across four additional experiments, and finally describe the implementation details.

2.1 Main Experiment

Valuing standard lotteries with price lists. Participants value binary lotteries $(\$x, p; \$y)$, which pay $\$x$ with probability p and $\$y$ otherwise. To study the fourfold pattern of risk attitudes (see, e.g., Wakker, 2010), we set $x \in \{25, -25\}$, $y = 0$, and $p \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. We denote these lotteries by GP and LP, where G (L) indicates a gain (loss) of \$25 and P is the probability in percent (e.g., G10 represents $(\$25, 0.10; \$0)$). We elicit valuations of these lotteries using price lists (see Figure 1). Each price list consists of 25 rows of choices, and each choice involves the fixed lottery option and a sure payment $\$c$ whose amount decreases monotonically by \$1 across rows, from \$25 to \$1 when $x = 25$ and from $-\$1$ to $-\$25$ when $x = -25$. Both options are represented by 100 boxes, each containing a specified amount, and the payment is the amount in a randomly selected box. For example, the lottery $(\$25, 0.10; \$0)$ is implemented by randomly opening one of 100

	Set A	Set B	
Question	100 Boxes	90 Boxes	10 Boxes
1	\$25.00	\$25	\$0
2	\$24.00	\$25	\$0
3	\$23.00	\$25	\$0
4	\$22.00	\$25	\$0
5	\$21.00	\$25	\$0
6	\$20.00	\$25	\$0
7	\$19.00	\$25	\$0
8	\$18.00	\$25	\$0

Figure 1: Partial screenshot of a price list for a standard lottery task (G90). *Notes:* The participant switches from the sure-amount option to the lottery option between \$21 and \$20, implying a certainty equivalent of \$20.50. The full list continues until the sure payment reaches \$1.

boxes, of which 10 contain \$25 and 90 contain \$0. For a sure payment $\$c$, all 100 boxes contain the same amount $\$c$.³

Test lotteries. We include eight test lotteries: two degenerate lotteries: $\pm(\$2.30, 0.10; \$2.30)$ and $\pm(\$22.70, 0.90; \$22.70)$; and two almost degenerate lotteries: $\pm(\$2.90, 0.10; \$2.10)$ and $\pm(\$22.90, 0.90; \$22.10)$. As with the standard lotteries, we elicit valuations of these test lotteries using price lists. Because the amounts in the sure payment are integers and the payoffs of the test lotteries lie between two consecutive integers, the row at which participants switch from the sure payment to the test lottery is unique and objectively determined by first-order stochastic dominance (FOSD). For example, for the test lottery $(\$2.90, 0.10; \$2.10)$, any participant who respects FOSD should switch to the lottery once the sure payment is \$2 or less, irrespective of their risk attitudes or their cognition in valuing lotteries. Table A.1 summarizes the stimuli.

³To cover possible losses, subjects receive an endowment of \$5 (gain lotteries) or \$30 (loss lotteries), applied identically to the standard and test lotteries.

2.2 Additional Experiments

Deterministic mirror. Building on the main experiment, the deterministic-mirror experiment includes an additional task of valuing riskless mirrors of lotteries. As proposed by Oprea (2024), each lottery presented in Table A.1, including the test lotteries, is replaced by its deterministic mirror in which payment is determined by opening *all* 100 boxes and paying the average amount, rather than by randomly opening one box, thereby eliminating risk. See Figure G.1 in the Appendix for an example screenshot of this task. In total, the mirror experiment contains 10 (8) price lists for standard (test) lotteries, and 10 (8) price lists for mirrors of standard (test) lotteries.

Cognitive uncertainty. The cognitive-uncertainty experiment uses the same stimuli as the main experiment and additionally elicits participants’ cognitive uncertainty about their choices in the price lists, following Enke and Graeber (2023). After submitting a response to each price list, participants are reminded of their choice and asked to indicate how certain they feel about it using a slider. Figure G.2 in the Appendix shows an example screenshot of this interface. We elicit cognitive uncertainty for both standard and test lotteries.

Student sample. In the student-sample experiment, we implement essentially the same design as in the main experiment, with two differences: (i) the sample consists of students from Radboud University in the Netherlands, rather than Prolific participants; and (ii) participants face stronger monetary incentives, as all participants — rather than only a subset — receive a bonus whose amount depends on one randomly selected choice, in accordance with the university laboratory regulations.

Binary choice. In the binary-choice experiment, we use the same standard and test lotteries as in the main experiment, with two differences. First, participants face binary choices, each on a separate decision screen, rather than price lists. Second, for each lottery we include only two binary choices: the lottery against the two sure amounts nearest its expected value (e.g., $\pm(\$2.90, 0.10; \$2.10)$ versus $\pm\$2$ and versus $\pm\$3$). Figure G.3 shows a screenshot of this binary-choice interface.

2.3 Implementation

We targeted 150 participants per experiment, recruited via Prolific from a U.S.-representative sample. The final valid samples were 150 for the main and deterministic-mirror experiments, 148 for the cognitive-uncertainty experiment, and 154 for the binary-choice experiment. The sample was gender-balanced and restricted to participants aged 18–75 with a historical approval rate of at least 99%. For incentives, in addition to a \$5 participation fee, each participant had a 10% chance of being selected for payment based on one of their choices made in the experiment. In the student-sample experiment, we recruited 81 participants. In addition to a €5 participation fee, all participants received a bonus determined by one randomly selected choice, up to a maximum of €30.

In experiments involving price lists, after reading the experimental instructions and before valuing lotteries with price lists, participants completed a short training module in which they saw examples illustrating the lotteries and the price lists. They then answered several comprehension questions. In the main experiment, there were six comprehension questions: three about the payment options and three about the price lists (see Appendix G). In the deterministic-mirror experiment, we added three questions about the payment options of mirrors. In the cognitive-uncertainty experiment, we included one additional comprehension question about cognitive uncertainty. In the binary-choice experiment, there were only three comprehension questions about the payment options. To encourage effort, participants received an additional \$0.15 for answering each comprehension question correctly on the first attempt. Participants who failed on the first attempt needed to try again and were required to answer all comprehension questions correctly before proceeding to the valuation task.

In experiments involving price lists, to reduce fatigue and boredom, participants were asked to indicate the row at which they would switch from the lottery (or the mirror) to the sure payment. When participants clicked a row in the lottery or mirror column, the lotteries/mirrors in that row and all rows below were highlighted. Analogously, clicking a row in the sure payment column highlighted the sure payment choice in that row and all rows above. Participants could revise their switching row as often as they wished and

finalized their decision by clicking the “Submit” button. Appendix G.1 presents a complete screenshot of the decision screen, along with additional screenshots of the experimental interface. In the binary-choice experiment, participants simply chose their preferred option in each binary choice.

In the four price-list experiments, the order of price lists was randomized separately within the gain and loss domains. At the end of the price-list task, we repeated two price lists. In one list, the lottery was either $(\$25, 0.10; \$0)$ or $(\$25, 0.90; \$0)$; in the other list, the lottery was either $(-\$25, 0.10; \$0)$ or $(-\$25, 0.90; \$0)$. The same logic applied to the binary-choice experiment.⁴ After completing the main choice task, participants reported how complex they found the task. They then faced eight non-incentivized binary choices, each on a separate decision screen. Following these, we elicited participants’ general risk preferences and administered three cognitive-reflection questions. At the end of the experiment, participants completed a short demographic questionnaire (gender, age, education, etc.). Figure A.1 illustrates the sequence of tasks in the main experiment. The experimental sequences for the additional experiments are reported in the corresponding appendices. All experimental protocols were approved by the Institutional Review Board at the National University of Singapore.

3 Experimental results

We present the experimental results from the main experiment and four additional experiments: the deterministic-mirror, cognitive-uncertainty, student-sample, and binary-choice experiments. The first two additional experiments incorporate the notions of deterministic mirror and cognitive uncertainty into the main experiment. The last two additional experiments replace the Prolific sample with a student sample, as well as price lists with binary choices.

⁴As in the price-list experiments, the binary choices were randomized separately within the gain and loss domains, and four repeated choices were appended to each payoff domain.

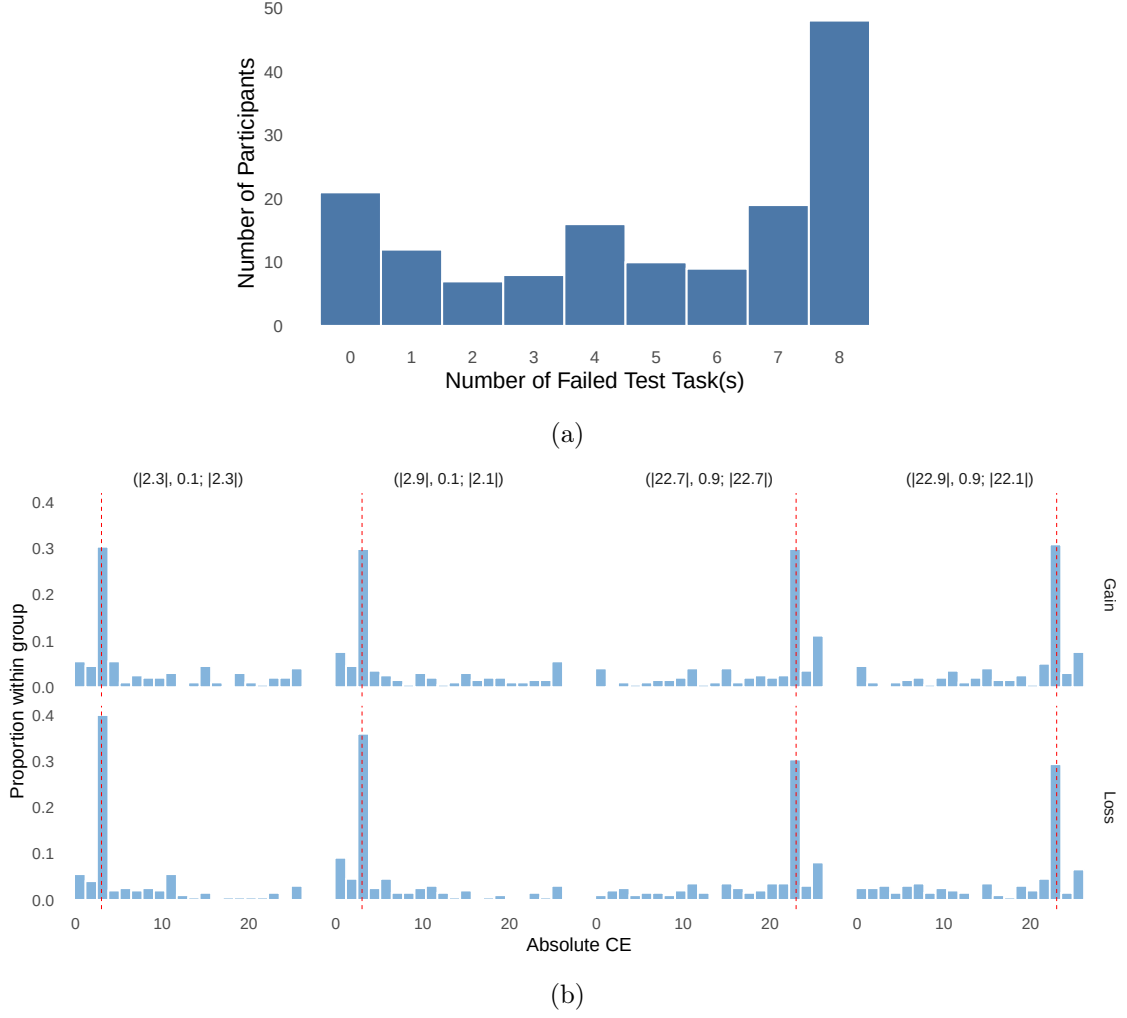


Figure 2: Performance on the ADLE test. *Notes:* Panel (a) shows the distribution of the number of ADLE tasks failed by each participant. Panel (b) plots the distributions of CEs across participants in each of the eight ADLE tasks. Red dashed lines denote the correct CE for each test lottery. The sample consists of 150 participants.

3.1 Main Experiment

In the main experiment and the other three experiments using price lists, we define a lottery's certainty equivalent (CE) as the average of the sure payments in the two switching rows. For example, if a participant switches between \$9 and \$8, we set the lottery's CE as \$8.5. Recall that the valuation of a single test lottery is an *ADLE task*, and a participant fails an ADLE task if their CE of the test lottery differs from its correct value; a participant fails the *ADLE test* if they fail at least one ADLE task.

Observation 1A. *The majority of participants fail at least one ADLE task. When participants fail an ADLE task, their CEs for the test lottery are close to the correct values.*

Panel (a) of Figure 2 summarizes the number of ADLE tasks each participant fails. For each ADLE task, about 30%–40% of participants pass it. On average, 60.0% of participants fail the ADLE test with degenerate lotteries and 62.5% fail the ADLE test with almost degenerate lotteries, with no statistically significant difference between the two ($p > 0.1$).⁵ Aggregating across all eight ADLE tasks, only 21 out of 150 (14%) participants pass all ADLE tasks. Among the remaining 129 participants, 12 fail exactly one task, 7 fail exactly two, and 110 fail three or more.

Figure 2b depicts the distribution of CEs for the eight test lotteries across 150 participants. When participants fail an ADLE task, the majority of CEs cluster near the correct value, with frequent deviations on both sides. This pattern indicates that many participants roughly know the correct CE, though not perfectly. Importantly, there is no clear tendency for participants to report the middle value of the price list’s sure payments as their CE: among the participants who fail an ADLE task, only 2% have a CE for the test lottery equal to the middle value of the price list. Additionally, there is no clear pattern of broader forms of middle bias: only 22.8% of participants switch between the true value and the middle value.

Observation 1B. *Participants exhibit a fourfold pattern of risk attitudes, with the pattern substantially more pronounced among those who fail the ADLE test than among those who pass.*

Figure 3a illustrates the fourfold pattern of risk attitudes among all participants. The x -axis is the payout probability p in $(\$25, p; \$0)$ and $(-\$25, p; \$0)$, and the y -axis is the deviation of the CE from the expected value, with positive values indicating risk seeking and negative values indicating risk aversion. Participants on average exhibit a clear fourfold pattern: they are risk seeking when lotteries have low-probability gains (G10 and G25) or high-probability losses (L75 and L90), and they are risk averse when lotteries have

⁵Event splitting effects would predict more ADLE task failures with almost degenerate lotteries than with degenerate lotteries (see e.g., Figure 6 Panel B in Bernheim and Sprenger, 2020). The absence of such a pattern suggests that event splitting effects, if present, are weak in our experiment.

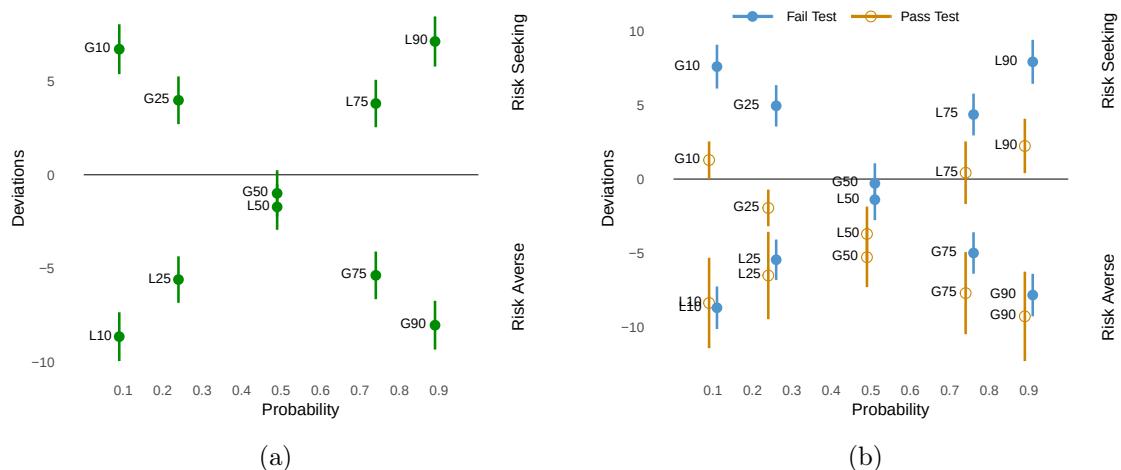


Figure 3: The fourfold pattern of risk attitudes and ADLE test performance. *Notes:* Panel (a) plots the fourfold pattern of risk attitudes for all participants. Panel (b) plots the same pattern separately for participants who pass versus those who fail the ADLE test. In both panels, the x -axis is the payout probability p in the lotteries $(\$25, p; \$0)$ and $(-\$25, p; \$0)$, and the y -axis is the deviation of the CE from the expected value, with positive values indicating risk seeking and negative values indicating risk aversion. Error bars correspond to ± 2 standard errors. Appendix Table A.2 reports the corresponding summary statistics of Panel (b).

high-probability gains (G75 and G90) or low-probability losses (L10 and L25). This figure replicates the classical findings in the literature (see, e.g., Kahneman and Tversky, 1979; Tversky and Kahneman, 1992).

Figure 3b plots the fourfold pattern separately for participants who pass the ADLE test versus those who fail. The figure reveals a clear pattern: the fourfold pattern is substantially more pronounced among participants who fail the ADLE test than among those who pass, with the largest differences arising in low-probability gains and high-probability losses.⁶

To complement this graphical evidence, we estimate a series of mixed-effects regressions of lotteries' CEs on the lotteries' payout probability p within each domain. For better comparison across domains, we take the normalized absolute value of the CEs, $\frac{|CE|}{25}$. The regression results in Columns (2) and (6) of Table 1 corroborate the patterns in Figure 3b: the coefficient on the interaction between payout probability p and ADLE test performance dummy D_T is significantly negative. This implies that participants who fail the ADLE test

⁶We do not claim that the fourfold pattern among participants who pass the ADLE test reflects what would be observed if all participants fully understood the decision environment. Participants who fail the ADLE test may have lower cognitive capacity. With additional instruction and practice, such participants may pass the test but still exhibit a pronounced fourfold pattern (Choi et al., 2022). In this sense, the fourfold pattern among participants who pass the ADLE test may understate the fourfold pattern.

Table 1: Mixed-effects regressions of normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.247*** (0.030)	0.479*** (0.079)	0.314*** (0.040)	0.495*** (0.079)	0.209*** (0.030)	0.451*** (0.078)	0.224*** (0.040)	0.449*** (0.079)
D_T		0.314*** (0.067)		0.251*** (0.067)		0.039 (0.068)		0.047 (0.070)
$p \times D_T$		-0.270** (0.085)		-0.230** (0.088)		-0.281*** (0.085)		-0.286** (0.087)
$D_{Compreh}$			0.223*** (0.046)	0.180*** (0.047)			-0.015 (0.048)	-0.023 (0.049)
$p \times D_{Compreh}$			-0.153* (0.060)	-0.114 (0.061)			-0.034 (0.060)	0.014 (0.061)
Intercept	0.347*** (0.024)	0.077 (0.062)	0.249*** (0.031)	0.051 (0.061)	0.436*** (0.024)	0.402*** (0.063)	0.442*** (0.032)	0.406*** (0.063)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; $D_T = 1$ indicates participants who failed the ADLE test; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

are significantly less sensitive to changes in likelihood, which is consistent with a stronger fourfold pattern.

We conduct several analyses to examine the robustness of these findings. First, we replace the performance dummy variable D_T with the continuous variable F_T , which records the number of failed ADLE tasks (see Table A.3). Second, we estimate Tobit regressions to account for the two-sided censored CEs between 0 and 25 (see Table A.4). Our observations remain essentially unchanged.⁷

Observation 1C. *ADLE test performance cannot be fully accounted for by performance on comprehension questions, and its association with fourfold risk attitudes remains robust to controlling for comprehension performance.*

While performance on the ADLE test and the comprehension questions are significantly correlated (Spearman's $\rho = 0.242, p < 0.001$), performance on comprehension questions does not fully account for performance on the ADLE test. Among the 66 participants who fail the comprehension questions on their first attempt, 95% (63 participants) fail the

⁷Beyond exhibiting a stronger fourfold pattern, participants who fail the ADLE test also make more preference reversals between their price-list choices and the corresponding binary choices, and are less consistent across repeated price lists (see Tables F.5, F.6, and F.7).

ADLE test (see Figure A.2 in the Appendix). While this proportion is significantly lower among participants who pass the comprehension questions (79% versus 95%; two-sided proportional test, $p < 0.01$), the majority in this group (66 out of 84) still fail the ADLE test.

To examine this more closely, we estimate mixed-effects regressions of $|CE_{ij}|/25$ on the comprehension dummy $D_{\text{Compreh},i}$ for each payoff domain, which equals 1 if participant i fails to answer all comprehension questions correctly on the first attempt. The results in Columns (3) and (7) of Table 1 suggest that performance on comprehension questions is systematically related to the strength of the fourfold pattern: participants who answer all comprehension questions correctly exhibit greater sensitivity to changes in payout probability than those who fail at least one question. More importantly, when we include both $D_{T,i}$ and $D_{\text{Compreh},i}$ in the mixed-effects regressions, the coefficients associated with D_T remain stable relative to the baseline specification without comprehension controls (Columns (2) versus (4) for gains and Columns (6) versus (8) for losses). By contrast, once D_T is included, the coefficients on D_{Compreh} become weaker and are often statistically insignificant (Columns (4) versus (3) for gains and Columns (8) versus (7) for losses). These results are robust across alternative specifications (see Tables A.3 and A.4 in the Appendix). Overall, these results indicate that comprehension questions do correlate with behavior but are less effective than the ADLE test at screening for behavioral heterogeneity in the fourfold risk attitudes.

3.2 Additional Experiment – Deterministic Mirror

We refer to the proportion of boxes containing \$25 or -\$25 in mirrors as the *payout proportion*, to distinguish it from the payout probability of lotteries. We define CEs as before and define the *simplicity equivalent* (SE) as the value of the mirror, following Oprea (2024). Unless explicitly stated otherwise, we classify participants as passing (failing) the ADLE test based on their valuation of the test lotteries, and use ADLE test performance on the mirrors only when stated explicitly. The deterministic-mirror experiment replicates the key findings from the main experiment. To avoid duplication, the corresponding figures and tables are reported in Appendix B; here we focus on the new results involving behavioral

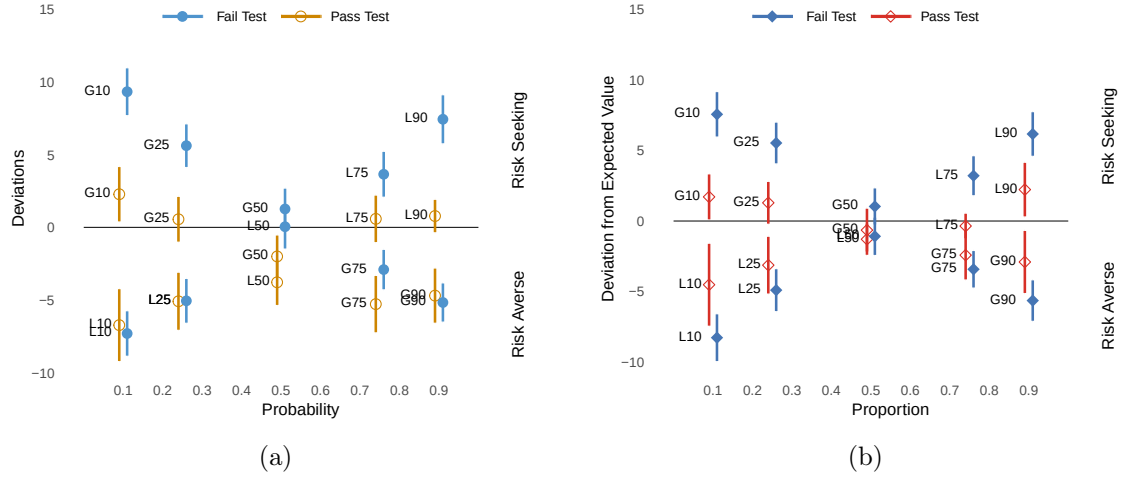


Figure 4: Comparisons of the fourfold pattern across lotteries and mirrors. *Notes:* Panel (a) plots deviations of CEs from expected value for lotteries, separately for participants who pass the ADLE test versus those who fail. Panel (b) plots deviations of SEs from the average amount for mirrors, separately for participants who pass the mirror ADLE test versus those who fail. Error bars correspond to ± 2 standard errors. See Appendix Tables B.1 for corresponding summary statistics of Panel (a), and B.2 for that of Panel (b).

patterns in the mirrors and their correspondence with those in the lotteries.

Observation 2A. *In both lotteries and mirrors, the fourfold pattern is stronger among participants who fail the ADLE test than among those who pass.*

Consistent with Oprea (2024), behavior in the deterministic-mirror experiment exhibits a fourfold pattern not only for lotteries but also for mirrors (see Figure B.3 in the Appendix). Figures 4a and 4b plot, respectively, deviations of lotteries' CEs from expected values by payout probability and deviations of mirrors' SEs from the average amount by payout proportion, with participants split by whether they pass or fail the ADLE test in the corresponding task. The fourfold pattern is clearly stronger among those who fail the ADLE test than among those who pass the test.⁸ This not only reinforces our finding in the main experiment but also provides a complementary perspective: participants who pass the ADLE test are significantly more responsive in their SEs to changes in mirror payout proportions.

Observation 2B. *Valuations of lotteries and mirrors differ significantly among participants who pass the ADLE test, but not among those who fail.*

⁸These results are robust to the order in which tasks were presented (see Figure B.4 in Appendix B).

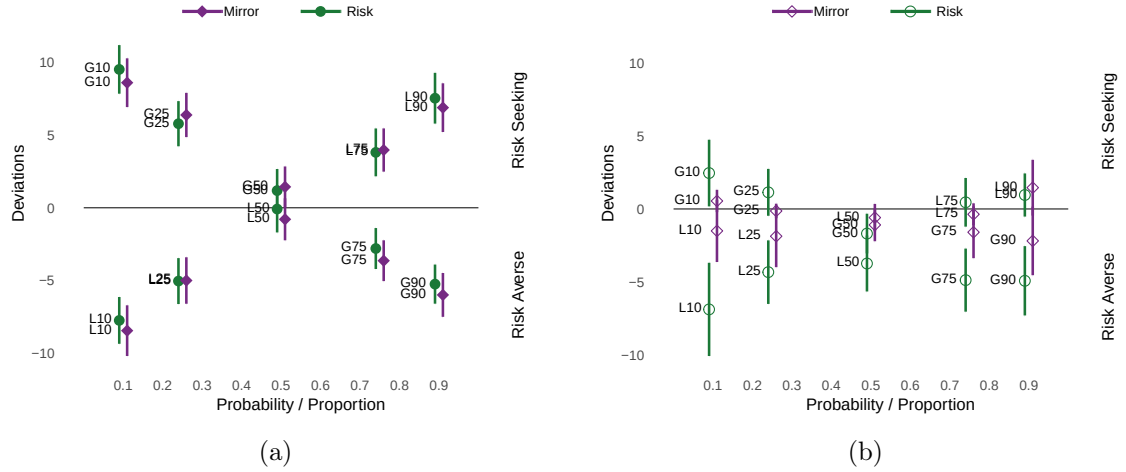


Figure 5: Comparisons of the fourfold pattern across lotteries and mirrors. *Notes:* Panel (a) plots deviations of CE (lotteries) against deviations of SE (mirrors) for participants who fail the ADLE test in both tasks. Panel (b) plots the same comparison for participants who pass the ADLE test in both tasks. Error bars correspond to ± 2 standard errors. See Appendix Tables B.3 for corresponding summary statistics of Panel (a), and B.4 for that of Panel (b).

For each lottery and its mirror (i.e., same domain and with lottery probability equal to mirror payout proportion), deviations of CEs from expected value are significantly correlated with the corresponding deviations of SEs from the average amount (Spearman’s $\rho = 0.472$, $p < 0.01$), consistent with Oprea (2024). Figure 5a further shows that, among participants who fail the ADLE test in both tasks, deviations in lotteries and mirrors move closely together: Spearman’s $\rho = 0.522$, $p < 0.01$, and these deviations do not differ significantly (Wilcoxon signed-rank test, $p > 0.10$). By contrast, among participants who pass the ADLE test in both tasks (Figure 5b), there is no clear relationship between the two types of deviations: Spearman’s $\rho = 0.138$, $p > 0.10$, and the deviations differ significantly between lotteries and mirrors (Wilcoxon signed-rank test, $p < 0.01$). These results suggest that the similarity between lotteries and mirrors is unlikely to be due to the computational complexity of the lotteries. The following observation provides further evidence for this interpretation.

Observation 2C. *After controlling for performance on the mirror tasks, the relationship between the performance on the ADLE test and the fourfold pattern in lotteries remains significant and strong.*

The mirrors also involve an objectively correct valuation, but arriving at it requires an

Table 2: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.356*** (0.027)	0.607*** (0.055)	0.697*** (0.096)	0.697*** (0.095)	0.342*** (0.028)	0.592*** (0.057)	0.657*** (0.100)	0.657*** (0.099)
D_T		0.303*** (0.050)		0.266*** (0.059)		0.060 (0.054)		0.041 (0.064)
$p \times D_T$		-0.331*** (0.063)		-0.285*** (0.074)		-0.328*** (0.065)		-0.296*** (0.077)
D_{mirror}			0.333*** (0.081)	0.113 (0.093)			0.091 (0.087)	0.057 (0.101)
$p \times D_{\text{mirror}}$			-0.371*** (0.100)	-0.136 (0.116)			-0.342** (0.104)	-0.098 (0.121)
Intercept	0.354*** (0.022)	0.123** (0.044)	0.048 (0.078)	0.048 (0.076)	0.364*** (0.024)	0.318*** (0.047)	0.280*** (0.083)	0.280*** (0.082)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{\text{mirror}} = 1$ indicates participants whose SE deviates the mirror's objective value at least once. Standard errors in parentheses. Columns (1–4) report gain domain results; columns (5–8) report loss domain results. All models use random effects with participant-level clustering.

extra step of calculation — participants must compute the average payoff across all 100 boxes — introducing a layer of lottery-specific cognitive friction beyond what the ADLE test captures. Among the 36 participants who pass the lottery ADLE test, 24 fail at least one mirror task. Among those who fail the lottery ADLE test, all also fail at least one mirror task.

We estimate a series of mixed-effects models of CEs on payout probability that include a dummy variable $D_{\text{mirror},i}$, which equals 1 if participant i fails at least one mirror task. Columns (3) and (7) of Table 2 indicate that participants who fail the mirror tasks tend to exhibit a stronger fourfold pattern in lotteries. However, when we extend the model to include the test dummy D_T , the coefficients associated with D_{mirror} are no longer statistically significant in either domain (compare Columns (3) and (7) with Columns (4) and (8)). By contrast, the coefficients associated with D_T remain essentially unchanged in both magnitude and significance when D_{mirror} is added to the model. This pattern is robust to several alternative specifications: replacing the dummy variable D_{mirror} with the continuous SE of the corresponding mirror whose payout proportion equals the lottery's probability, and also in Tobit models (see Tables B.5 and B.6). Together, these results indicate that, when options are simple, the association between mirror performance and

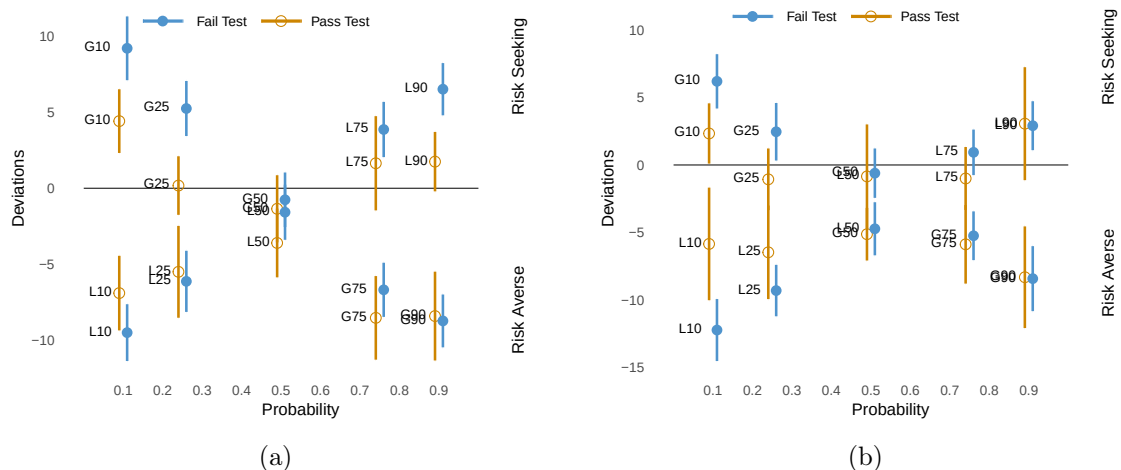


Figure 6: Comparison of the fourfold pattern by cognitive uncertainty. *Notes:* Panel (a) plots deviations of lotteries' CEs from expected values for high-CU choices, separately for participants who pass and who fail the ADLE test. Panel (b) plots the corresponding deviations for low-CU choices. Error bars correspond to ± 2 standard errors. See Appendix Tables C.1 for corresponding summary statistics of Panel (a), and C.2 for that of Panel (b).

the fourfold pattern largely reflects the underlying relationship between ADLE test performance and the fourfold pattern, with the additional computational friction of the mirror task playing a limited independent role.

3.3 Additional Experiment – Cognitive Uncertainty

Observation 3A. *The relationship between ADLE test performance and the fourfold pattern remains robust to controls for cognitive uncertainty.*

Following Enke and Graeber (2023), we divide all risky choices into two groups: those with CU below the overall median (low-CU choices) and those with CU above the median (high-CU choices). Within each subsample, Figures 6a and 6b plot deviations of lotteries' CEs from expected value for participants who pass the ADLE test versus those who fail.⁹ For high-CU choices (Figure 6a), participants who fail the ADLE test display a markedly stronger fourfold pattern than those who pass. For low-CU choices (Figure 6b), the difference between the two performance groups is weaker but still visible, especially for the lotteries involving small probabilities.

⁹We again replicate the key findings from the main experiment. To avoid duplication, we report the full set of figures and tables in Appendix C and focus here on how the relationship between ADLE test performance and risky choice patterns varies across CU levels.

Table 3: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.236*** (0.025)	0.424*** (0.052)	0.348*** (0.044)	0.512*** (0.061)	0.271*** (0.025)	0.532*** (0.052)	0.337*** (0.044)	0.582*** (0.063)
D_T		0.221*** (0.051)		0.208*** (0.051)		0.188*** (0.053)		0.178*** (0.051)
$p \times D_T$		-0.241*** (0.059)		-0.223*** (0.059)		-0.337*** (0.059)		-0.313*** (0.059)
CU			0.370*** (0.091)	0.337*** (0.090)			-0.192* (0.080)	-0.164* (0.079)
$p \times CU$			-0.450** (0.144)	-0.411** (0.143)			-0.223 (0.115)	-0.228* (0.112)
Intercept	0.331*** (0.022)	0.159*** (0.045)	0.239*** (0.031)	0.086 (0.049)	0.479*** (0.022)	0.333*** (0.047)	0.542*** (0.034)	0.394*** (0.052)
Observations	740	740	740	740	740	740	740	740

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; D_T is a dummy variable (1 = failed the ADLE test, 0 = passed); CU_{ij} indicates cognitive uncertainty measure for lottery j of participant i . Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

To supplement this graphical evidence, we estimate mixed-effects regressions analogous to those in the main experiment, but now include a continuous variable CU_{ij} , which measures participant i 's CU for lottery j 's CE. Table 3 reports the results. Columns (1)–(2) and (5)–(6) replicate the core patterns in the main experiment and the deterministic-mirror experiment: failing the ADLE test is associated with greater likelihood insensitivity, which is consistent with a stronger fourfold pattern. Columns (3) and (7) show that higher CU choices are less sensitive to changes in payout probability than lower CU choices, consistent with the “compression effects” visible in Figures 6a and 6b. Most importantly, when we include both $D_{T,i}$ and CU_{ij} (and their interactions with p_{ij}) in the regression, the coefficients on $D_{T,i}$ and its interaction change very little relative to the specifications without CU. This indicates that the link between likelihood insensitivity and ADLE test performance is robust to controlling for CU. These conclusions remain essentially unchanged when we replace the continuous CU measure with an individual CU dummy or when we estimate Tobit models instead of mixed-effects linear models (see Tables C.3 and C.4).

Observation 3B. *Even after controlling for uncertainty about standard lotteries, the fourfold pattern is strongly linked to cognitive uncertainty about the test lotteries.*

Table 4: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>			<i>Loss Domain</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
p	0.296*** (0.035)	0.268*** (0.051)	0.267*** (0.051)	0.367*** (0.035)	0.402*** (0.051)	0.400*** (0.051)
CU^{test}	0.421*** (0.127)		0.565*** (0.153)	0.017 (0.128)		0.132 (0.154)
$p \times CU^{\text{test}}$	-0.360* (0.147)		-0.440* (0.178)	-0.583*** (0.147)		-0.493** (0.178)
CU^{risk}		0.069 (0.142)	-0.283 (0.169)		-0.142 (0.141)	-0.224 (0.170)
$p \times CU^{\text{risk}}$		-0.117 (0.164)	0.157 (0.197)		-0.484** (0.164)	-0.177 (0.198)
Intercept	0.261*** (0.030)	0.312*** (0.044)	0.314*** (0.044)	0.476*** (0.030)	0.518*** (0.044)	0.518*** (0.044)
Observations	740	740	740	740	740	740

Notes: Dependent variable = normalized absolute certainty equivalent. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. CU^{test} represents participant i 's median value of test-list cognitive uncertainty; CU^{risk} represents participant i 's median value of risk-lottery-list cognitive uncertainty. Standard errors in parentheses. All models use random effects with participant-level clustering.

We further compare CU measures for the standard and test lotteries, and assess their relative importance in the fourfold pattern. On average, CU for the standard lotteries correlates significantly with CU for the test lotteries (Spearman's $\rho = 0.576$, $p < 0.01$). However, this correlation is strong and significant among participants who fail the ADLE test (Spearman's $\rho = 0.744$, $p < 0.01$) but not among those who pass (Spearman's $\rho = 0.064$, $p > 0.10$). Since the test lotteries are straightforward to evaluate, these results suggest that CU for lotteries can arise from factors other than valuation difficulty. For participants who pass the ADLE test, CU reflects the inherent difficulty of valuing the lotteries, whereas for participants who fail the ADLE test, CU primarily arises from factors beyond valuation difficulty, consistent with our earlier findings on the deterministic mirrors (see Table C.5 in Appendix C for additional evidence).

To gain deeper insight into the relative influence of the two types of CU — one stemming from valuation and the other from broader contextual factors — we estimate a series of mixed-effects regressions of normalized absolute certainty equivalents, $\frac{|CE_{ij}|}{25}$, on two individual-level average CU variables and their interaction terms with the payout probability. The first variable CU_{ij}^{risk} represents the median level of participant i 's average CU

for the standard lotteries, while CU_{ij}^{test} is defined analogously for the test lotteries. The regression results are reported in Table 4.

By comparing Columns (1) and (4) with Columns (2) and (5), we find that individual median CU for the test lotteries is strongly associated with risk lottery’s certainty equivalents, while individual median CU for the risk lotteries is less so. More importantly, when both CU variables and their interaction terms with the payout probability are included in the regression (Columns (3) and (6)), the coefficients associated with CU for the test lotteries remain largely unchanged compared to specifications (1) and (4), whereas the coefficients related to CU for the standard lotteries lose statistical significance. This result indicates that, in our simple setting with binary lotteries, CU that is not specific to lottery valuation contributes to the fourfold pattern, while the additional layer of CU driven by the lottery valuation appears to be less significant.¹⁰

3.4 Additional Experiment – Student Sample

Observation 4A. *More than one-third of student participants fail the ADLE test.*

Relative to the Prolific sample, the student participants perform significantly better on the ADLE test. As shown in Figure 7, a substantially higher proportion of students pass the ADLE test (52 out of 81 in the student sample, versus 21 out of 150 in the Prolific sample). Among those who fail the ADLE test, many students fail only once (18 out of 29 in the student sample, compared to 12 out of 129 in the Prolific sample in the main experiment). Figure D.1 in the Appendix presents histograms of ADLE test performance for both samples, illustrating this difference. This result is consistent with Snowberg and Yariv (2021), who compare choice behavior across subject pools (a student population, a representative U.S. sample, and Amazon Mechanical Turk subjects) and find lower noise in the student sample.

Observation 4B. *The fourfold pattern is significantly stronger among student participants who fail the ADLE test than among those who pass. This pattern is robust to controlling*

¹⁰We also replace two individual median CU values with two CU dummies indicating whether the participants’ median CU is higher than the group median level, and the main findings remain robust. See Table C.6 for further details.

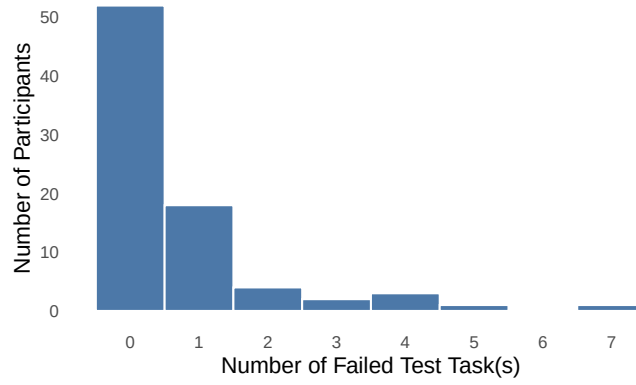


Figure 7: ADLE test performance in students.

for performance on comprehension questions.

Figure 8 presents the fourfold pattern for the student participants and its comparison between two ADLE test performance groups. In the left panel, the fourfold pattern is significant, consistent with previous similar studies with a student population (e.g., Bruhin et al., 2010; L’Haridon and Vieider, 2019). In the right panel, we split the sample into two groups based on their ADLE test performance. Again, student participants who fail the ADLE test exhibit a stronger fourfold pattern than those who pass, broadly consistent with those from the main experiment, although the differences between the two groups are somewhat weaker. This smaller difference likely reflects the fact that many student participants fail the ADLE test only once, as noted above.

Finally, Table 5 presents mixed-effects regressions. Results suggest that the fourfold pattern is stronger among participants who fail the ADLE test, and the relationship between ADLE test performance and the fourfold pattern is robust to the inclusion of comprehension performance, similar to the main experiment (see related robustness checks in Appendix D).

3.5 Additional Experiment - Binary choice

In the binary-choice experiment, each standard or test lottery is compared against the integer sure amounts just below and just above its expected value as two binary choices. To keep ADLE test performance in the binary-choice experiment comparable with the

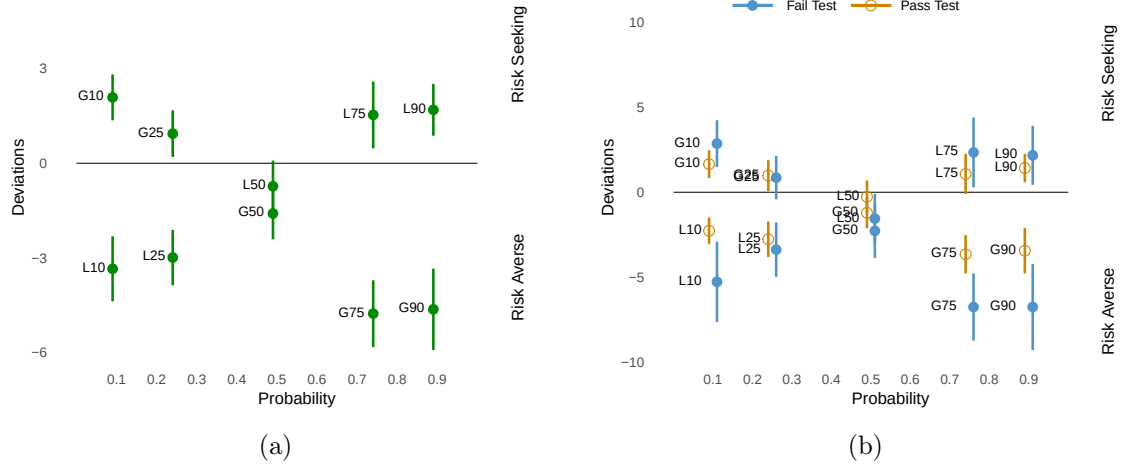


Figure 8: The fourfold pattern of risk attitudes and ADLE test performance (student sample). *Notes:* Panel (a) plots the fourfold pattern of risk attitudes for all participants. Panel (b) plots the same pattern separately for participants who pass the ADLE test and those who fail. In both panels, the x -axis is the payout probability p in the lotteries $(\$25, p; \$0)$ and $(-\$25, p; \$0)$, and the y -axis is the deviation of the CE from the expected value, with positive values indicating risk seeking and negative values indicating risk aversion. Error bars correspond to ± 2 standard errors. Appendix Table D.1 reports the corresponding summary statistics.

Table 5: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.619*** (0.024)	0.701*** (0.029)	0.617*** (0.026)	0.694*** (0.029)	0.706*** (0.023)	0.770*** (0.028)	0.713*** (0.025)	0.770*** (0.029)
D_T		0.064 (0.036)		0.069 (0.036)		0.112** (0.035)		0.120*** (0.036)
$p \times D_T$		-0.230*** (0.048)		-0.241*** (0.049)		-0.177*** (0.047)		-0.177*** (0.048)
$D_{Compreh}$			-0.016 (0.049)	-0.034 (0.049)			-0.024 (0.048)	-0.056 (0.048)
$p \times D_{Compreh}$			0.014 (0.067)	0.077 (0.066)			-0.047 (0.064)	0.000 (0.064)
Intercept	0.127*** (0.018)	0.104*** (0.021)	0.129*** (0.019)	0.107*** (0.022)	0.178*** (0.017)	0.138*** (0.021)	0.181*** (0.018)	0.143*** (0.022)
Observations	405	405	405	405	405	405	405	405

Notes: Dependent variable = $|CE|/25$ (normalized absolute certainty equivalent). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Standard errors in parentheses. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

price-list experiments, where each test lottery constitutes a single ADLE task, we treat the two binary choices for a test lottery as one test choice pair: a participant passes the pair only if she chooses the test lottery over the lower amount and the higher amount over the test lottery (e.g., for $(\$2.90, 0.10; \$2.10)$, choosing the test lottery over \$2 and \$3 over

the test lottery) — and fails the ADLE test if she fails at least one pair. For the standard lotteries, we classify a participant as risk seeking for a given lottery if she chooses the lottery in both binary choices, that is, over both the sure payments below and above its expected value.

Observation 5A. *Compared to the main experiment, participants in the binary-choice experiment are substantially less likely to fail the ADLE test and also less likely to exhibit the fourfold pattern.*

We find that 40.3% of participants (62 out of 154) pass all eight ADLE test choice pairs, and only 3.9% (6) of participants fail all of them, compared with 14.0% and 32.0%, respectively, in the main experiment (two-proportion z-tests, $p < 0.001$ for both comparisons). Figure E.1 in Appendix E further plots the distribution of the number of test failures in the two experiments; failure rates are substantially lower in the binary-choice experiment than in the main experiment. This is consistent with the significantly lower perceived complexity in the binary-choice experiment than that in the main experiment (1.85 versus 2.39, Wilcoxon rank sum test, $p < 0.01$).

Figure 9a reports the proportion of risk-seeking responses for each standard lottery in the main experiment and the binary-choice experiment. In contrast to the main experiment, the majority of choices are risk averse across all probabilities and both payoff domains in the binary-choice experiment. This finding is consistent with the results reported by Harbaugh et al. (2010) and Bouchouicha et al. (2026).

Observation 5B. *In the binary-choice experiment, participants who fail the ADLE test do not exhibit a stronger fourfold pattern than those who pass.*

Figure 9b plots the proportion of risk-seeking participants in the binary-choice experiment, separately by ADLE test performance. Unlike in the price-list experiments, behavior differs little by test performance. If anything, participants who pass the ADLE test exhibit somewhat stronger probability dependence, whereas those who fail display a flatter pattern, with the proportion classified as risk seeking varying little across probabilities. Table 6 examines this relationship using mixed-effects logit regressions in which the dependent variable equals one if the participant chooses the lottery in both binary choices for

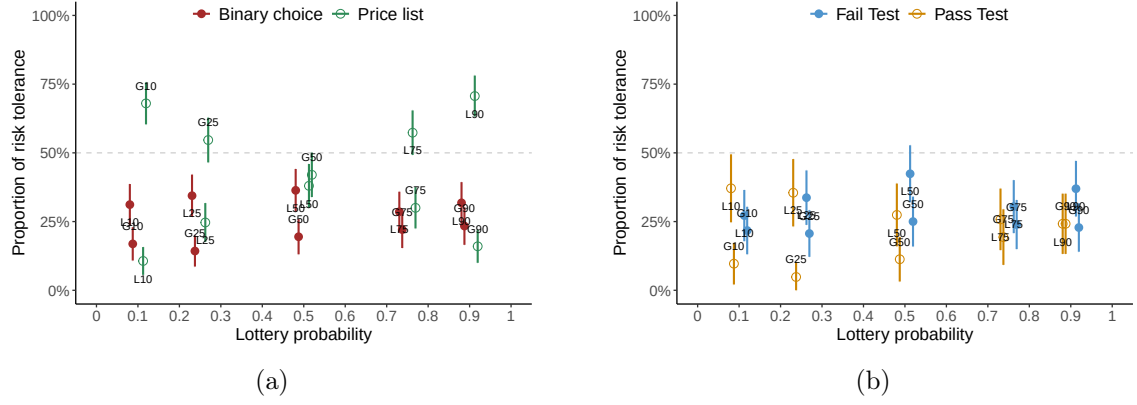


Figure 9: Risk seeking in the binary-choice and price-list experiments. *Notes:* Panel (a) plots, for each standard lottery, the proportion of risk-seeking participants in the price-list and binary-choice experiments. Panel (b) plots the same proportion in the binary-choice experiment separately by ADLE test performance. A participant is classified as risk seeking for a lottery if they choose the lottery in both binary choices bracketing its expected value (binary-choice experiment) or if their certainty equivalent exceeds the lottery's expected value (price-list experiment). Error bars show 95% normal-approximation confidence intervals, and the dashed horizontal line denotes a proportion of 50%.

Table 6: Mixed-effects logit regressions of risk seeking in standard binary choices

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.863* (0.356)	2.325*** (0.679)	0.600 (0.420)	2.146** (0.687)	-0.165 (0.292)	-1.087* (0.488)	-0.361 (0.345)	-1.131* (0.495)
D_T		1.974*** (0.587)		2.089*** (0.605)		-0.512 (0.425)		-0.404 (0.439)
$p \times D_T$		-2.113** (0.802)		-2.514** (0.835)		1.511* (0.624)		1.422* (0.643)
D_{Compreh}			0.068 (0.571)	-0.474 (0.593)			-0.587 (0.493)	-0.480 (0.512)
$p \times D_{\text{Compreh}}$			0.929 (0.792)	1.571 (0.831)			0.812 (0.705)	0.417 (0.730)
Intercept	-2.436*** (0.284)	-3.768*** (0.534)	-2.448*** (0.325)	-3.723*** (0.538)	-0.912*** (0.206)	-0.614 (0.327)	-0.778** (0.240)	-0.564 (0.331)
Observations	770	770	770	770	770	770	770	770
Participants	154	154	154	154	154	154	154	154

Notes: The dependent variable equals one if participant i is classified as risk seeking for a given standard lottery — that is, chooses the lottery in both binary choices bracketing its expected value — and zero otherwise. p is the lottery's payout probability. $D_T = 1$ indicates participants who failed the ADLE test. $D_{\text{Compreh}} = 1$ indicates participants who failed to answer all three comprehension questions correctly on the first attempt. All models are mixed-effects logit regressions with participant random intercepts. Standard errors are in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

a given lottery — that is, is classified as risk seeking. In the gain domain, a higher winning probability significantly increases the likelihood of being risk seeking among participants who pass the test, but this probability dependence is substantially weaker among those who fail (Column (2)). The loss domain shows the mirror-image pattern, with risk seek-

ing declining in probability among passers and again much flatter among those who fail (Column (6)). Finally, consistent with the price-list experiments, the relationship between ADLE test performance and risk taking remains robust after controlling for comprehension, as shown in Columns (4) and (8).

4 Conclusion and Discussion

To understand cognitive frictions in decisions under risk, we introduce a test based on almost degenerate lottery evaluation (ADLE). The ADLE test is a model-agnostic diagnostic, as valuations should converge to the degenerate value under any model that respects dominance. Across the experiments using price lists, the majority of participants fail the ADLE test, and the failure is systematically linked to a stronger fourfold pattern, even after controlling for performance on comprehension questions, behavior in the deterministic mirror task, and the degree of cognitive uncertainty. We further identify key boundary conditions for these results: both the failure rate and the magnitude of the association are attenuated when examining a university student sample rather than the Prolific sample, and when replacing price-list elicitation with binary-choice tasks. The remainder of this section discusses the broader theoretical and methodological implications of our results.

Lottery-specific valuation. Most preference-based models predict that the certainty equivalent of an almost degenerate lottery converges to that of its degenerate value. Consequently, these frameworks cannot account for our findings. Nevertheless, presenting lotteries with two (almost) identical outcomes may induce event-splitting effects (Starmer and Sugden, 1993; Bernheim and Sprenger, 2020), which can generate violations of dominance. Some theories may accommodate such violations.¹¹ For example, in the original prospect theory (Kahneman and Tversky, 1979), dominance violations can arise under subcertainty ($w(p) + w(1 - p) < 1$), in which an almost degenerate lottery is valued strictly below its degenerate counterpart. Crucially, since dominance violations depend on $w(p) + w(1 - p)$ rather than $w(p)$ alone, the usual association between probability weighting and the four-

¹¹In addition to the original prospect theory (Kahneman and Tversky, 1979), preference for gambling (Fishburn, 1980; Diecidue et al., 2004), the uncertainty effect (Gneezy et al., 2006), and complexity aversion (Puri, 2025; de Clippel et al., 2025) may also lead to dominance violation

fold pattern may not carry over to ADLE task failures.

Our experimental data are inconsistent with these predictions. First, we find that the average CE of $(\$2.90, 0.10; \$2.10)$ is above 2.9 and the average CE of $(\$22.90, 0.90; \$22.10)$ is below 22.10. Note that subcertainty in the original prospect theory can account for the latter result but not the former. Second, failure rates on the ADLE task are virtually indistinguishable between degenerate and almost degenerate lotteries. Most importantly, ADLE test performance is systematically associated with the fourfold pattern. Taken together, these findings suggest that existing mechanisms accommodating dominance violations are unlikely to play a central role in our setting. Our findings point toward cognitive frictions inherent in the decision environment rather than lottery-specific perception and valuation.

Cognition in decision environments. Going beyond lottery-specific perception and valuation, our findings thus point to various sources of cognitive friction in the decision environment. For instance, elicitation mechanisms such as the Becker–DeGroot–Marschak (BDM) mechanism can be cognitively demanding and cause confusion (Plott and Zeiler, 2005, 2007; Cason and Plott, 2014). Much of the methodology literature emphasizes giving clear mechanism descriptions (Gonczarowski et al., 2023), reducing misconceptions through training (Plott and Zeiler, 2005, 2007), improving instructions (Freeman et al., 2018), and adopting a characterization assessment method (Bernheim et al., 2025). We use price-list elicitation in the first four experiments, which is arguably simpler than the BDM mechanism. Moreover, we include both examples and comprehension questions about how the price lists operate, and we require participants to answer all comprehension questions correctly before proceeding to the price lists. Nevertheless, performance on comprehension questions does not appear to be strongly linked to the ADLE test.

Furthermore, our findings may be partially influenced by the inattention of participants. Since the experiments typically involve low stakes, participants may not feel motivated to engage fully, resulting in limited cognitive effort. This is consistent with the different passing rates of the ADLE test between the Prolific and student samples, even though our experimental compensation aligns with standard hourly wages in both samples. It also echoes the observation in Snowberg and Yariv (2021) that noise tends to be higher in online samples than in student samples.

Cognitive frictions arising from the choice environment can often be mitigated by strengthening incentives, heightening cognitive engagement, or simplifying the elicitation procedure. Consistent with this intuition, participants in our binary-choice experiment are substantially less likely both to fail the ADLE test and to exhibit the fourfold pattern relative to the baseline experiment. The failure might be reduced further when participants are notified of dominance violations, prompted to reconsider their decisions, or presented with larger stake sizes (Benjamin et al., 2020; Nielsen and Rehbeck, 2022; Gneezy et al., 2024). Disentangling these distinct cognitive frictions represents a fruitful avenue for future research.

Heuristics and Noise. When participants confront their cognitive constraints, they may adopt heuristics or make random decisions (Simon, 1955; Tversky and Kahneman, 1974; Gigerenzer et al., 2000). A prominent heuristic under risk is middle bias — the tendency to switch in the middle of price lists or to select the midpoint in budgetary settings (Read and Loewenstein, 1995; Benartzi and Thaler, 2001; Rubinstein, 2002; Choi et al., 2006; Enke and Graeber, 2023; Halevy and Mayraz, 2024). Individuals may also make noisy decisions by randomizing between options, deliberately or otherwise (Machina, 1985; Hey, 2005; Agranov and Ortoleva, 2017; Cettolin and Riedl, 2019; Halevy et al., 2023).¹² In the context of price lists, both middle bias and random decisions can generate likelihood insensitivity.

The distribution of certainty equivalents (CEs) for (almost) degenerate lotteries may help us understand the underlying heuristics and noise. A pronounced form of middle bias suggests that participants tend to switch their choices near the midpoint of the price list. However, we observe very few instances of this behavior. A milder form of middle bias is that participants may anchor their valuations around the midpoint and insufficiently adjust toward the correct value, that is, they switch between the correct value and the midpoint. While a substantial portion of CEs (22.8%) falls within this range, a larger portion (36.2%) lies outside it. We leave it to future studies to explore the distinction between heuristics and noise, as well as to examine their implications for likelihood insensitivity and related

¹²Other heuristic rules include the status quo bias (Masatlioglu and Ok, 2005; Ortoleva, 2010), simplicity seeking (Iyengar and Kamenica, 2010), caution (Cerreia-Vioglio et al., 2015; Gilboa et al., 2021; Cerreia-Vioglio et al., 2024), and mental accounting (Gilboa et al., 2021).

anomalies.

Together, our findings demonstrate that canonical risk anomalies can emerge directly from cognitive frictions inherent in the decision environment. More broadly, our work highlights how the multifaceted cognitive demands of choice architecture can blur the boundary between underlying preferences, heuristics and noise.

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Appendices

A Appendix: Additional results in the main experiment

Table A.1: Options in the price lists.

Price lists	Lotteries	The sure payment
Standard lotteries	$\pm(\$25, 0.10; \$0)$, $\pm(\$25, 0.25; \$0)$, $\pm(\$25, 0.50; \$0)$, $\pm(\$25, 0.75; \$0)$, $\pm(\$25, 0.90; \$0)$	From \$25 to \$1 for gain lists and -\$1 to -\$25 for loss lists in increments of \$1.
Test lotteries	$\pm(\$2.30, 0.10; \$2.30)$, $\pm(\$22.70, 0.90; \$22.70)$, $\pm(\$2.90, 0.10; \$2.10)$, $\pm(\$22.90, 0.90; \$22.10)$	

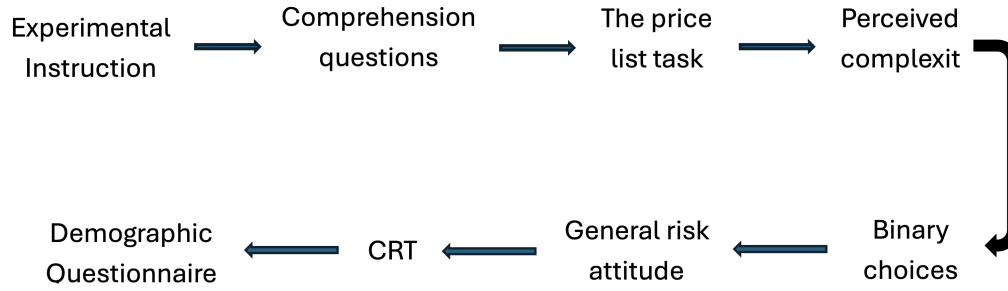


Figure A.1: Procedure for the main experiment.

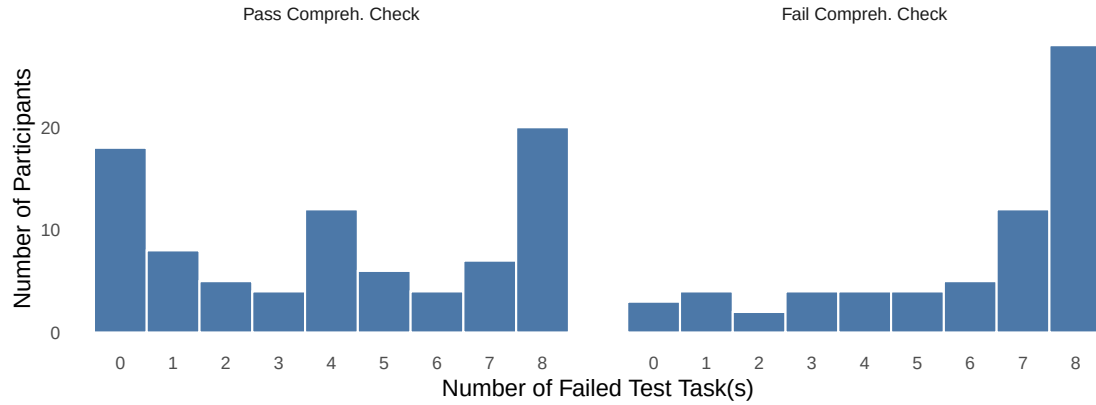


Figure A.2: Participants' performance on the ADLE test (the number of failed ADLE tasks).

Table A.2: Deviations of certainty equivalents from expected values by the ADLE test performance.

P	ADLE	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	21	3.786	2.5	0.054	0.049	0.001	0.002
10	1	G10	129	10.105	8.5	0.000	0.000		
25	0	G25	21	4.548	4.5	0.005	0.012	0.000	0.000
25	1	G25	129	11.453	9.5	0.000	0.000		
50	0	G50	21	7.214	5.5	0.000	0.000	0.005	0.007
50	1	G50	129	12.205	12.5	0.000	0.000		
75	0	G75	21	10.786	9.5	0.019	0.031	0.139	0.103
75	1	G75	129	13.500	14.5	0.000	0.000		
90	0	G90	21	13.214	11.5	0.000	0.002	0.445	0.316
90	1	G90	129	14.655	14.5	0.000	0.000		
Panel B: Loss domain									
10	0	L10	21	-10.881	-10.5	0.000	0.229	0.864	0.989
10	1	L10	129	-11.205	-10.5	0.000	0.162		
25	0	L25	21	-13.024	-12.5	0.000	0.137	0.553	0.469
25	1	L25	129	-11.958	-10.5	0.000	0.172		
50	0	L50	21	-16.214	-15.5	0.001	0.001	0.189	0.194
50	1	L50	129	-13.896	-14.5	0.046	0.003		
75	0	L75	21	-18.071	-19.5	0.689	0.878	0.031	0.047
75	1	L75	129	-14.136	-15.5	0.000	0.002		
90	0	L90	21	-20.262	-20.5	0.024	0.002	0.003	0.011
90	1	L90	129	-14.570	-15.5	0.000	0.000		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. All values rounded to three decimals. See corresponding Figure 3b.

Table A.3: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.247*** (0.030)	0.488*** (0.056)	0.290*** (0.036)	0.489*** (0.056)	0.209*** (0.030)	0.502*** (0.055)	0.245*** (0.035)	0.501*** (0.055)
F_T		0.053*** (0.008)		0.047*** (0.008)		0.007 (0.008)		0.008 (0.008)
$p \times F_T$		-0.049*** (0.010)		-0.047*** (0.010)		-0.060*** (0.010)		-0.061*** (0.010)
$F_{Compreh}$			0.084*** (0.019)	0.045* (0.020)			0.002 (0.019)	-0.005 (0.020)
$p \times F_{Compreh}$			-0.053* (0.024)	-0.014 (0.026)			-0.044 (0.024)	0.007 (0.025)
Intercept	0.347*** (0.024)	0.088* (0.043)	0.279*** (0.028)	0.082 (0.043)	0.436*** (0.024)	0.400*** (0.044)	0.434*** (0.028)	0.401*** (0.044)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; F_T indicates the number of ADLE tasks participants failed; $F_{Compreh}$ indicates the number of comprehension questions participants failed. Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

Table A.4: Mixed-effects Tobit regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.273*** (0.035)	0.515*** (0.091)	0.344*** (0.046)	0.532*** (0.092)	0.233*** (0.034)	0.472*** (0.089)	0.243*** (0.045)	0.468*** (0.090)
D_T		0.323*** (0.080)		0.250** (0.081)		0.045 (0.077)		0.053 (0.079)
$p \times D_T$		-0.283** (0.099)		-0.242* (0.101)		-0.279** (0.096)		-0.288** (0.099)
$D_{Compreh}$			0.258*** (0.056)	0.215*** (0.057)			-0.014 (0.054)	-0.023 (0.055)
$p \times D_{Compreh}$			-0.160* (0.070)	-0.118 (0.072)			-0.023 (0.069)	0.025 (0.070)
Intercept	0.324*** (0.029)	0.047 (0.074)	0.211*** (0.037)	0.016 (0.073)	0.424*** (0.027)	0.386*** (0.071)	0.430*** (0.036)	0.389*** (0.072)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE|/25$. Reported coefficients are from mixed-effects Tobit models with double censoring at 0.02 and 1.02. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE tests; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random intercepts.

B Appendix: Additional Results in the Mirror Experiment

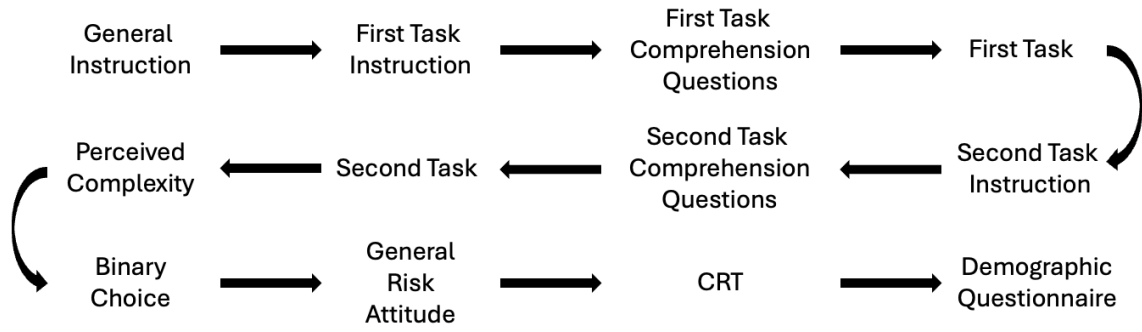


Figure B.1: Procedure for the deterministic-mirror experiment. The lottery task and the mirror task are presented in a randomized order, with one serving as the first task and the other as the second.

Table B.1: Deviations of certainty equivalents from expected values by the ADLE test performance.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	36	4.778	2.5	0.020	0.019	0.000	0.000
10	1	G10	114	11.825	9.5	0.000	0.000		
25	0	G25	36	7.056	6.5	0.474	0.626	0.000	0.000
25	1	G25	114	12.114	10.0	0.000	0.000		
50	0	G50	36	10.500	9.5	0.008	0.001	0.013	0.015
50	1	G50	114	13.763	12.5	0.073	0.101		
75	0	G75	36	13.222	14.5	0.000	0.000	0.074	0.053
75	1	G75	114	15.596	15.5	0.000	0.000		
90	0	G90	36	17.806	19.5	0.000	0.000	0.712	0.901
90	1	G90	114	17.333	19.5	0.000	0.000		
Panel B: Loss domain									
10	0	L10	36	-9.222	-8.0	0.000	0.000	0.706	0.856
10	1	L10	114	-9.798	-6.5	0.000	0.000		
25	0	L25	36	-11.583	-10.5	0.000	0.000	0.983	0.659
25	1	L25	114	-11.553	-10.5	0.000	0.000		
50	0	L50	36	-16.278	-15.5	0.000	0.000	0.007	0.007
50	1	L50	114	-12.456	-12.5	0.953	0.861		
75	0	L75	36	-17.917	-18.5	0.469	0.980	0.035	0.100
75	1	L75	114	-14.851	-15.5	0.000	0.000		
90	0	L90	36	-21.722	-22.5	0.169	0.484	0.000	0.001
90	1	L90	114	-15.070	-15.5	0.000	0.000		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. T_{fail} = indicator for ADLE failure (1 = fail, 0 = pass). All values rounded to three decimals.

Table B.2: Deviations of simplicity equivalents from expected values by the ADLE test performance.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	31	4.210	2.5	0.040	0.022	0.000	0.000
10	1	G10	119	10.063	6.5	0.000	0.000		
25	0	G25	31	7.790	6.5	0.091	0.091	0.004	0.011
25	1	G25	119	12.029	9.5	0.000	0.000		
50	0	G50	31	11.855	12.5	0.401	0.288	0.202	0.205
50	1	G50	119	13.534	12.5	0.107	0.140		
75	0	G75	31	16.081	18.5	0.008	0.011	0.456	0.674
75	1	G75	119	15.080	17.5	0.000	0.000		
90	0	G90	31	19.597	22.5	0.013	0.012	0.072	0.224
90	1	G90	119	16.861	19.5	0.000	0.000		
Panel B: Loss domain									
10	0	L10	31	-7.016	-2.5	0.004	0.006	0.037	0.078
10	1	L10	119	-10.769	-8.5	0.000	0.000		
25	0	L25	31	-9.629	-6.5	0.004	0.009	0.252	0.462
25	1	L25	119	-11.399	-8.5	0.000	0.000		
50	0	L50	31	-13.790	-12.5	0.026	0.022	0.876	0.718
50	1	L50	119	-13.584	-12.5	0.102	0.080		
75	0	L75	31	-18.855	-18.5	0.419	0.561	0.010	0.062
75	1	L75	119	-15.290	-18.5	0.000	0.000		
90	0	L90	31	-20.274	-22.5	0.025	0.012	0.014	0.067
90	1	L90	119	-16.332	-20.5	0.000	0.000		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across ADLE-test groups. W2 = Wilcoxon rank-sum test across ADLE-test groups. T_{fail} = indicator for ADLE-test failure (1 = fail, 0 = pass). All values rounded to three decimals.

Table B.3: Deviations of certainty/simplicity equivalents from expected values among those who failed the ADLE test.

P	Treatment	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	mirror	G10	105	11.100	9.5	0.000	0.000	0.441	0.458
10	risk	G10	105	12.014	9.5	0.000	0.000		
25	mirror	G25	105	12.881	11.5	0.000	0.000	0.581	0.666
25	risk	G25	105	12.281	10.5	0.000	0.000		
50	mirror	G50	105	13.938	12.5	0.043	0.056	0.801	0.665
50	risk	G50	105	13.681	12.5	0.115	0.148		
75	mirror	G75	105	14.862	17.5	0.000	0.000	0.400	0.399
75	risk	G75	105	15.700	15.5	0.000	0.001		
90	mirror	G90	105	16.510	17.5	0.000	0.000	0.463	0.641
90	risk	G90	105	17.252	19.5	0.000	0.000		
Panel B: Loss domain									
10	mirror	L10	105	-10.948	-9.5	0.000	0.000	0.553	0.740
10	risk	L10	105	-10.243	-7.5	0.000	0.000		
25	mirror	L25	105	-11.500	-10.5	0.000	0.000	0.973	0.885
25	risk	L25	105	-11.538	-10.5	0.000	0.000		
50	mirror	L50	105	-13.290	-12.5	0.276	0.239	0.515	0.530
50	risk	L50	105	-12.586	-12.5	0.915	0.715		
75	mirror	L75	105	-14.529	-15.5	0.000	0.000	0.884	0.650
75	risk	L75	105	-14.690	-15.5	0.000	0.000		
90	mirror	L90	105	-15.614	-19.5	0.000	0.000	0.593	0.844
90	risk	L90	105	-14.967	-15.5	0.000	0.000		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across treatments. W2 = Wilcoxon rank-sum test across treatments. Treatment indicates whether the mirror or risk version of the task was used. All values rounded to three decimals.

Table B.4: Deviations of certainty/simplicity equivalents from the expected values among those who passed the ADLE test.

P	Treatment	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	mirror	G10	22	3.045	2.5	0.168	0.371	0.120	0.081
10	risk	G10	22	4.955	3.0	0.043	0.037		
25	mirror	G25	22	6.364	6.5	0.589	0.789	0.137	0.296
25	risk	G25	22	7.636	6.5	0.171	0.159		
50	mirror	G50	22	11.409	12.5	0.064	0.100	0.505	0.426
50	risk	G50	22	10.818	11.0	0.022	0.008		
75	mirror	G75	22	16.909	18.5	0.087	0.098	0.024	0.011
75	risk	G75	22	13.636	14.5	0.000	0.001		
90	mirror	G90	22	20.318	22.5	0.077	0.057	0.109	0.046
90	risk	G90	22	17.591	18.5	0.000	0.001		
Panel B: Loss domain									
10	mirror	L10	22	-4.000	-2.5	0.173	0.269	0.008	0.007
10	risk	L10	22	-9.364	-9.0	0.000	0.001		
25	mirror	L25	22	-8.364	-6.5	0.093	0.174	0.113	0.026
25	risk	L25	22	-10.818	-10.5	0.001	0.002		
50	mirror	L50	22	-13.091	-12.5	0.221	0.371	0.005	0.014
50	risk	L50	22	-16.227	-15.5	0.001	0.002		
75	mirror	L75	22	-18.864	-18.5	0.344	0.461	0.375	0.599
75	risk	L75	22	-18.045	-18.5	0.590	0.856		
90	mirror	L90	22	-21.045	-22.5	0.142	0.100	0.680	0.368
90	risk	L90	22	-21.545	-22.5	0.209	0.477		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across treatments. W2 = Wilcoxon rank-sum test across treatments. Treatment indicates whether the mirror or risk version of the task was used. All values rounded to three decimals.

Table B.5: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.356*** (0.027)	0.607*** (0.055)	0.352*** (0.059)	0.483*** (0.077)	0.342*** (0.028)	0.592*** (0.057)	0.110 (0.060)	0.323*** (0.084)
D_T		0.303*** (0.050)		0.212*** (0.051)		0.060 (0.054)		0.028 (0.053)
$p \times D_T$		-0.331*** (0.063)		-0.198** (0.067)		-0.328*** (0.065)		-0.231*** (0.069)
SE			0.336*** (0.057)	0.256*** (0.060)			0.073 (0.056)	0.050 (0.056)
$p \times SE$			-0.181 (0.094)	-0.109 (0.095)			0.282** (0.092)	0.242** (0.091)
Intercept	0.354*** (0.022)	0.123** (0.044)	0.233*** (0.031)	0.102* (0.044)	0.364*** (0.024)	0.318*** (0.047)	0.354*** (0.033)	0.339*** (0.050)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; D_T is a dummy variable (1 = failed the ADLE test, 0 = passed); SE indicates the simplicity equivalents of the corresponding mirror question (percent = proportion). Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

Table B.6: Mixed-effects Tobit regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.389*** (0.031)	0.643*** (0.062)	0.736*** (0.109)	0.735*** (0.107)	0.372*** (0.032)	0.601*** (0.063)	0.657*** (0.110)	0.657*** (0.109)
D_T		0.319*** (0.058)		0.282*** (0.068)		0.053 (0.062)		0.036 (0.073)
$p \times D_T$		-0.336*** (0.071)		-0.290*** (0.084)		-0.305*** (0.073)		-0.277** (0.085)
D_{mirror}			0.344*** (0.093)	0.112 (0.107)			0.081 (0.097)	0.052 (0.113)
$p \times D_{\text{mirror}}$			-0.377*** (0.113)	-0.139 (0.131)			-0.311** (0.115)	-0.085 (0.133)
Intercept	0.336*** (0.026)	0.095 (0.051)	0.020 (0.089)	0.021 (0.087)	0.355*** (0.028)	0.315*** (0.054)	0.280** (0.092)	0.280** (0.091)
Observations	750	750	750	750	750	750	750	750

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE|/25$. Reported coefficients are from mixed-effects Tobit models with double censoring at 0.02 and 1.02. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{\text{mirror}} = 1$ indicates participants whose simplicity equivalent deviates from the mirror task's objective value at least once. Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random intercepts.

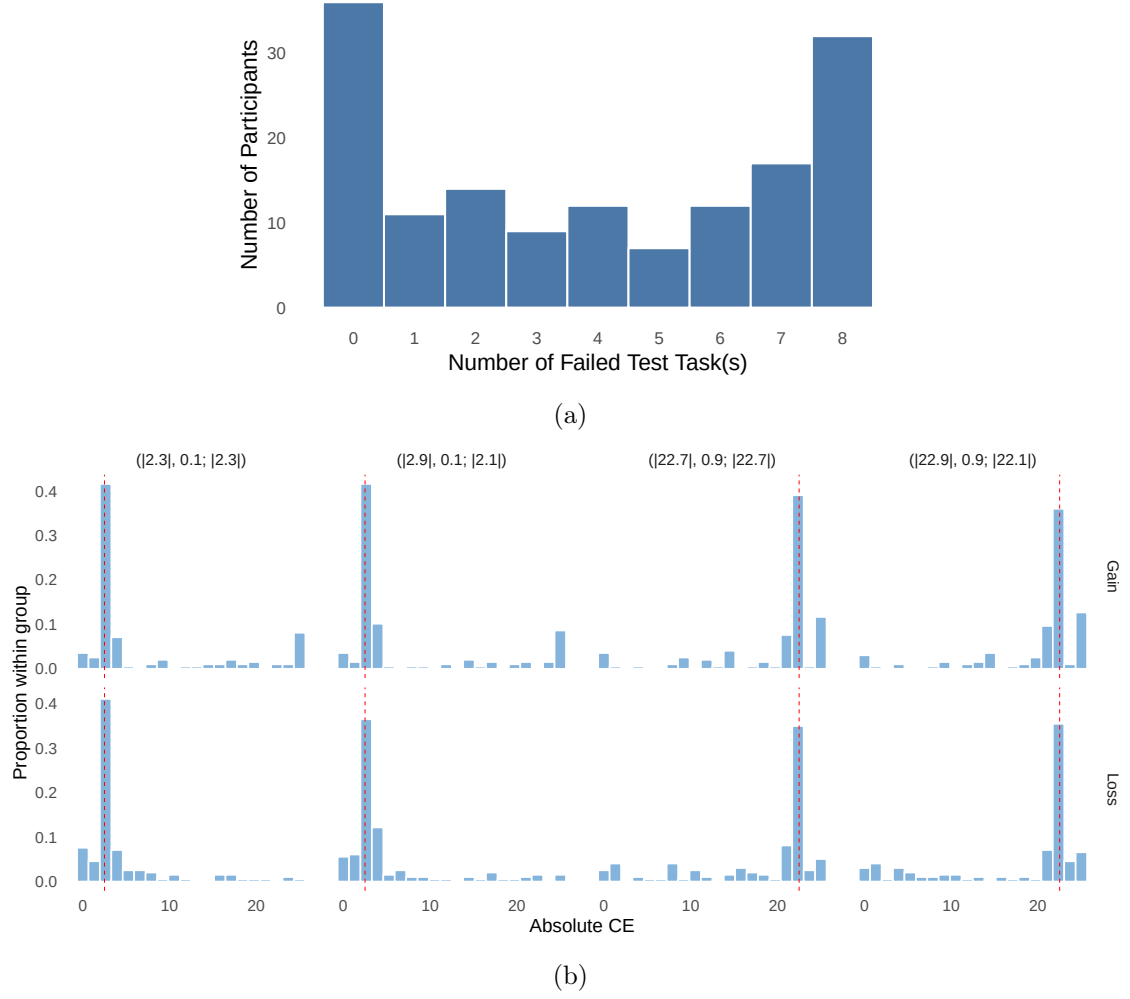


Figure B.2: Performance on the ADLE test. *Notes:* Panel (a) shows the distribution of the number of failed ADLE tasks by each participant. The sample consists of 150 participants. Panel (b) plots the distribution of individual CEs—i.e., switching positions—across the eight test lotteries, separately for gain and loss domains. Red dashed lines indicate the theoretically correct switching position for each list.

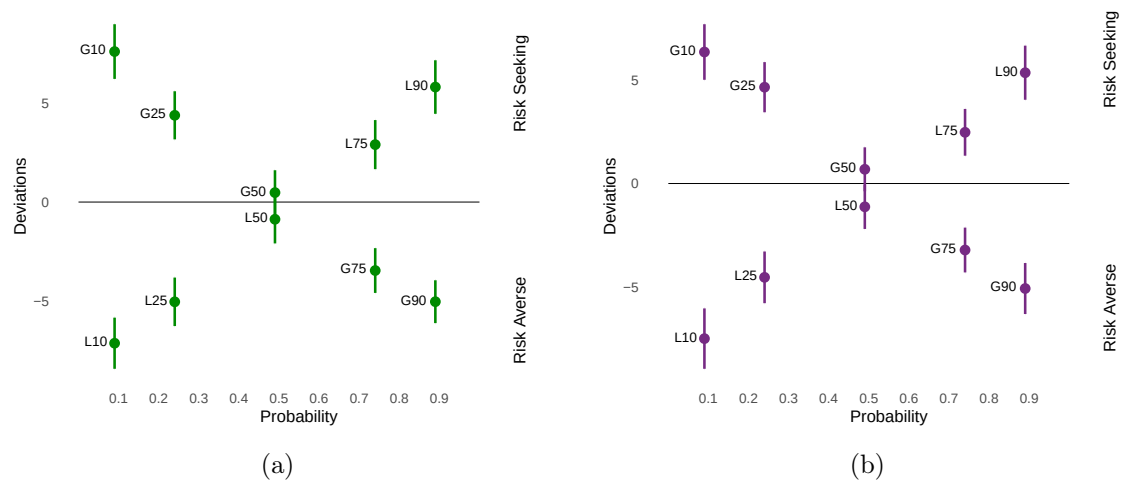
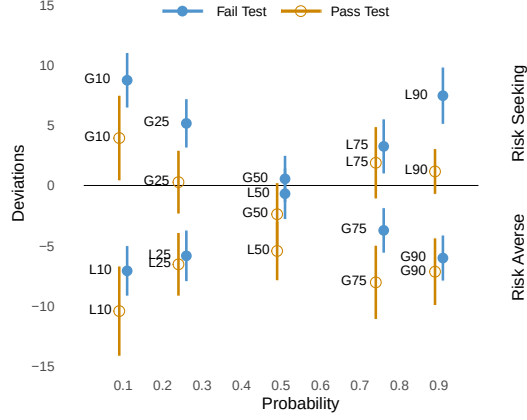
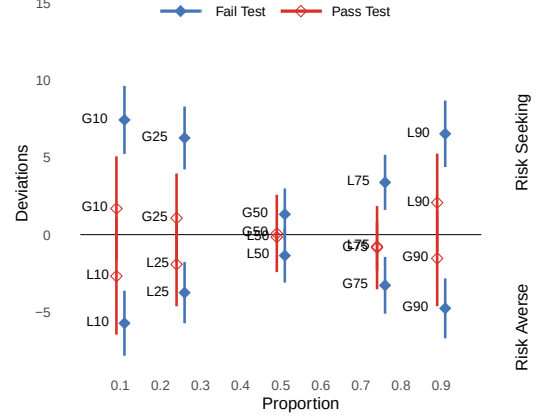


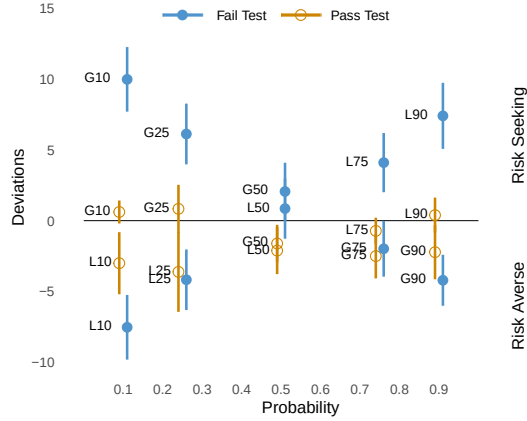
Figure B.3: Comparison of the fourfold patterns. *Notes:* Panel (a) represents the pattern of CE from lottery task; Panel (b) represents SE from mirror task.



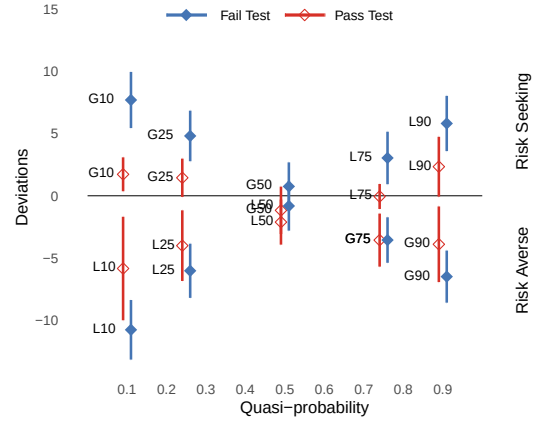
(a) First-answer lottery task response



(b) First-answer mirror task response



(c) Second-answer lottery task response



(d) Second-answer mirror task response

Figure B.4: The fourfold pattern comparison in first- and second-answer tasks for participants who pass or fail the ADLE test. Top row: first-answer tasks; bottom row: second-answer tasks. Left column: lottery task responses; right column: mirror task responses.

C Appendix: Additional results in Experiment CU

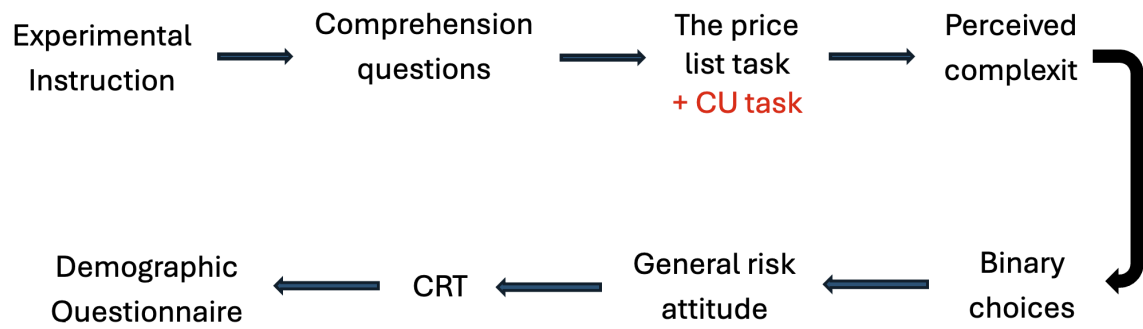


Figure C.1: Procedure for the cognitive-uncertainty experiment..

Table C.1: Deviations of certainty equivalents from expected values by the ADLE test performance among those of higher cognitive uncertainty.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	12	6.917	5.5	0.001	0.004	0.039	0.057
10	1	G10	53	11.700	9.5	0.000	0.000		
25	0	G25	17	6.676	5.5	0.860	0.776	0.008	0.018
25	1	G25	65	11.746	11.5	0.000	0.000		
50	0	G50	17	11.147	9.5	0.239	0.124	0.730	0.941
50	1	G50	54	11.740	10.5	0.401	0.301		
75	0	G75	17	9.971	9.5	0.001	0.000	0.308	0.287
75	1	G75	57	11.816	10.5	0.000	0.000		
90	0	G90	17	14.088	14.5	0.030	0.007	0.865	0.942
90	1	G90	67	13.769	14.5	0.000	0.000		
Panel B: Loss domain									
10	0	L10	20	-9.400	-10.5	0.000	0.000	0.129	0.177
10	1	L10	52	-12.000	-11.0	0.000	0.000		
25	0	L25	18	-11.200	-10.5	0.000	0.000	0.750	0.662
25	1	L25	55	-12.623	-11.5	0.000	0.000		
50	0	L50	20	-16.100	-15.5	0.005	0.007	0.231	0.229
50	1	L50	57	-14.079	-15.5	0.087	0.047		
75	0	L75	14	-16.857	-18.5	0.309	0.387	0.253	0.321
75	1	L75	52	-14.673	-15.5	0.000	0.000		
90	0	L90	16	-20.750	-21.0	0.093	0.120	0.008	0.010
90	1	L90	60	-15.983	-15.5	0.000	0.000		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. All values rounded to three decimals. See corresponding Figure 6a.

Table C.2: Deviations of certainty equivalents from expected values by the ADLE test performance among those of lower cognitive uncertainty.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	21	4.833	3.5	0.005	0.150	0.040	0.028
10	1	G10	62	8.694	6.0	0.000	0.000		
25	0	G25	16	5.438	4.5	0.367	0.163	0.083	0.089
25	1	G25	50	8.960	7.0	0.025	0.164		
50	0	G50	16	7.375	6.5	0.000	0.002	0.017	0.023
50	1	G50	61	11.893	10.5	0.509	0.388		
75	0	G75	16	12.625	14.5	0.001	0.003	0.737	0.812
75	1	G75	58	13.259	14.5	0.000	0.000		
90	0	G90	16	14.188	14.5	0.000	0.001	0.965	0.864
90	1	G90	48	14.083	13.0	0.000	0.000		
Panel B: Loss domain									
10	0	L10	13	-8.346	-4.5	0.016	0.014	0.021	0.014
10	1	L10	63	-14.722	-15.5	0.000	0.000		
25	0	L25	15	-12.967	-9.5	0.002	0.005	0.181	0.136
25	1	L25	60	-15.800	-16.0	0.000	0.000		
50	0	L50	13	-13.346	-14.5	0.680	0.592	0.092	0.072
50	1	L50	58	-17.224	-20.0	0.000	0.000		
75	0	L75	19	-19.500	-20.5	0.402	0.242	0.248	0.432
75	1	L75	63	-17.563	-19.5	0.271	0.278		
90	0	L90	17	-19.441	-22.5	0.163	0.843	0.940	0.841
90	1	L90	55	-19.591	-22.5	0.002	0.080		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. All values rounded to three decimals. See corresponding Figure 6b.

Table C.3: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.236*** (0.025)	0.424*** (0.052)	0.309*** (0.037)	0.482*** (0.057)	0.271*** (0.025)	0.532*** (0.052)	0.305*** (0.036)	0.566*** (0.059)
D_T		0.221*** (0.051)		0.213*** (0.051)		0.188*** (0.053)		0.183*** (0.052)
$p \times D_T$		-0.241*** (0.059)		-0.230*** (0.059)		-0.337*** (0.059)		-0.328*** (0.059)
CU_{high}			0.109*** (0.033)	0.101** (0.032)			-0.032 (0.034)	-0.015 (0.033)
$p \times CU_{\text{high}}$			-0.148** (0.053)	-0.136** (0.052)			-0.070 (0.053)	-0.083 (0.052)
Intercept	0.331*** (0.022)	0.159*** (0.045)	0.278*** (0.027)	0.116* (0.047)	0.479*** (0.022)	0.333*** (0.047)	0.495*** (0.027)	0.345*** (0.050)
Observations	740	740	740	740	740	740	740	740

Notes: Dependent variable = normalized absolute certainty equivalent. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $CU_{\text{high}} = 1$ indicates participant i 's cognitive uncertainty is higher than the median level of cognitive uncertainty for list j . Standard errors in parentheses. Columns (1–4) report gain domain results; columns (5–8) report loss domain results. All models use random effects with participant-level clustering.

Table C.4: Mixed-effects Tobit regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.249*** (0.027)	0.432*** (0.057)	0.379*** (0.048)	0.540*** (0.067)	0.294*** (0.028)	0.560*** (0.058)	0.371*** (0.050)	0.619*** (0.070)
D_T		0.227*** (0.056)		0.213*** (0.056)		0.210*** (0.058)		0.193*** (0.055)
$p \times D_T$		-0.237*** (0.065)		-0.219*** (0.064)		-0.344*** (0.066)		-0.319*** (0.065)
CU			0.407*** (0.100)	0.375*** (0.099)			-0.211* (0.089)	-0.185* (0.088)
$p \times CU$			-0.520*** (0.158)	-0.485** (0.156)			-0.257* (0.127)	-0.258* (0.125)
Intercept	0.320*** (0.024)	0.143** (0.050)	0.217*** (0.034)	0.060 (0.054)	0.478*** (0.025)	0.317*** (0.050)	0.545*** (0.037)	0.387*** (0.057)
Observations	740	740	740	740	740	740	740	740

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE|/25$. Reported coefficients are from mixed-effects Tobit models with double censoring at 0.02 and 1.02. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE tests; CU_{ij} denotes participant i 's CU for the certainty equivalent of lottery j . Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random intercepts.

Table C.5: Cognitive uncertainty (CU) about the test lotteries and the standard lotteries by the frequency of ADLE test failures.

	Overall	The frequency of ADLE test failures				
		0	1	2-4	5-7	8
Mean CU (Test lotteries)	0.185	0.080	0.104	0.165	0.261	0.330
Mean CU (Standard lotteries)	0.283	0.276	0.229	0.254	0.300	0.355
Difference in CU	0.098	0.196	0.125	0.089	0.039	0.025

Table C.6: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>			<i>Loss Domain</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
p	0.315*** (0.032)	0.269*** (0.036)	0.312*** (0.037)	0.360*** (0.032)	0.359*** (0.036)	0.393*** (0.037)
$CU_{\text{high}}^{\text{test}}$	0.180*** (0.044)		0.198*** (0.047)	0.044 (0.045)		0.038 (0.035)
$p \times CU_{\text{high}}^{\text{test}}$	-0.201*** (0.051)		-0.204*** (0.054)	-0.227*** (0.051)		-0.188*** (0.055)
$CU_{\text{high}}^{\text{risk}}$		0.025 (0.043)	-0.046 (0.046)		0.027 (0.044)	0.013 (0.047)
$p \times CU_{\text{high}}^{\text{risk}}$		-0.064 (0.050)	0.009 (0.053)		-0.171*** (0.050)	-0.104 (0.053)
Intercept	0.260*** (0.027)	0.318*** (0.031)	0.277*** (0.032)	0.462*** (0.028)	0.466*** (0.031)	0.458*** (0.033)
Observations	740	740	740	740	740	740

Notes: Dependent variable = normalized absolute certainty equivalent. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $CU_{\text{high}}^{\text{test}} = 1$ indicates participant i 's test-list cognitive uncertainty average is higher than the group median level; $CU_{\text{high}}^{\text{risk}} = 1$ indicates participant i 's risk-lottery-list cognitive uncertainty average is higher than the group median level. Standard errors in parentheses. All models use random effects with participant-level clustering.

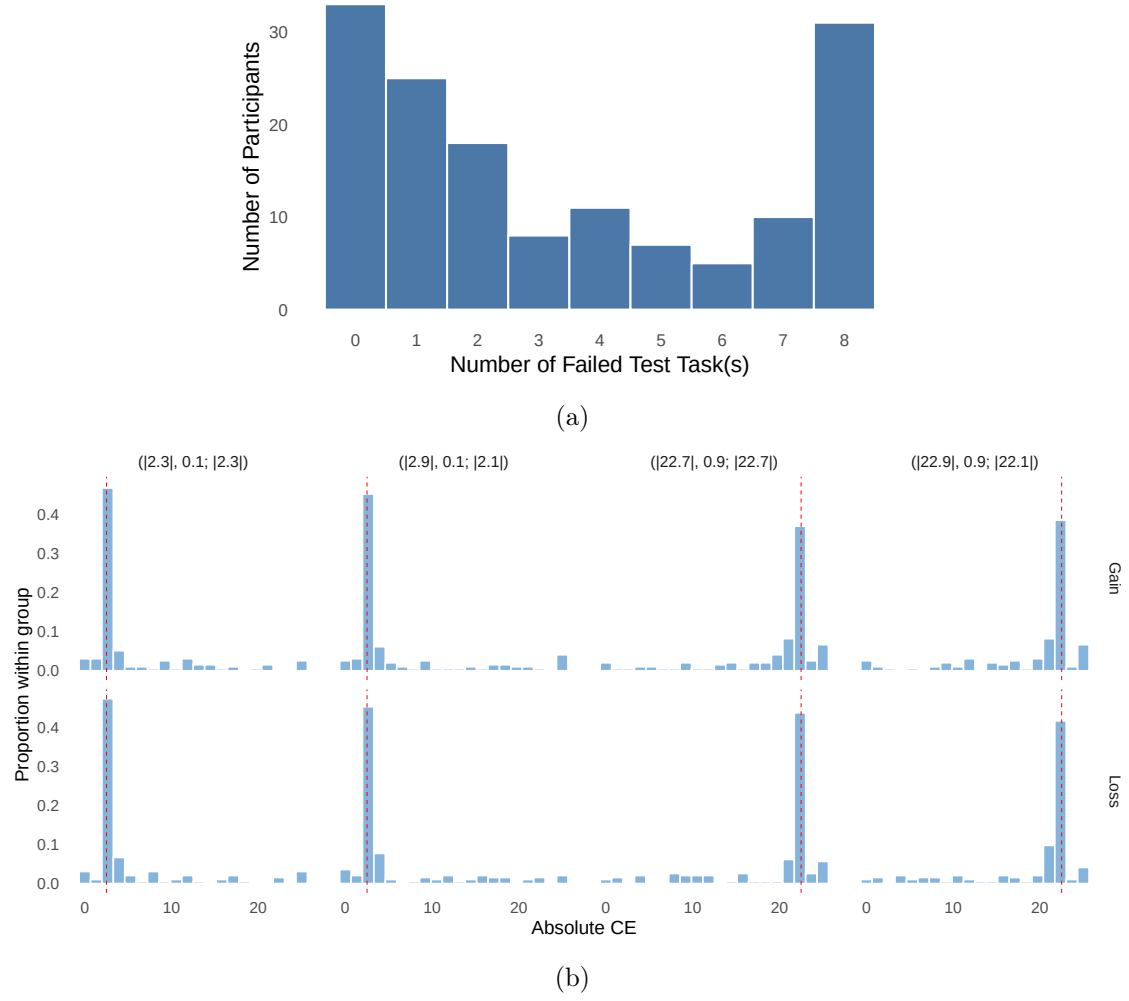


Figure C.2: Performance on the ADLE test. *Notes:* Panel (a) shows the distribution of the number of failed ADLE tasks by each participant. The sample consists of 148 participants. Panel (b) plots the distribution of individual CEs—i.e., switching positions—across the eight test lotteries, separately for gain and loss domains. Red dashed lines indicate the theoretically correct switching position for each list.

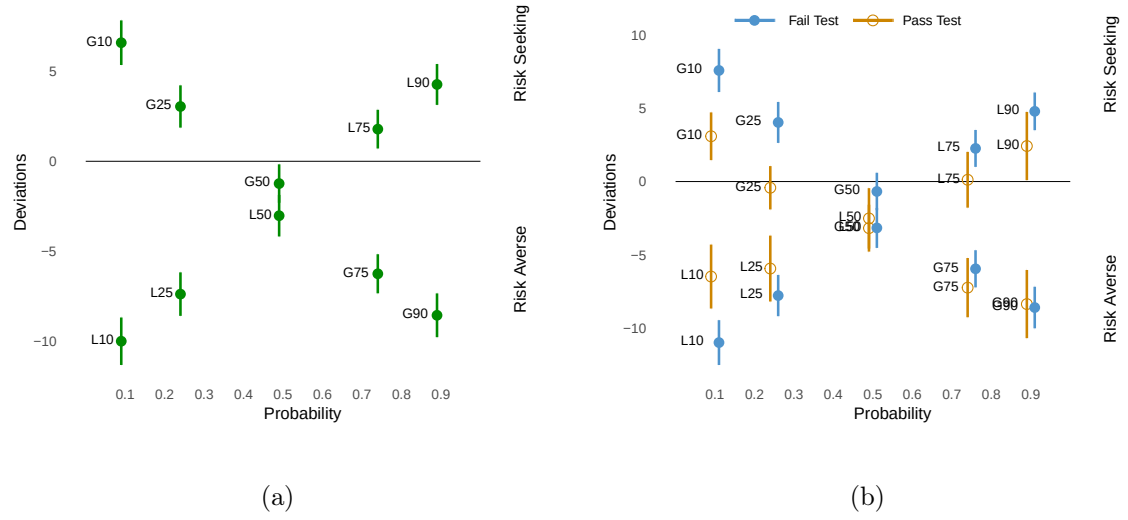


Figure C.3: The fourfold pattern of risk attitudes and ADLE test performance (cognitive-uncertainty experiment). *Notes:* Panel (a) plots the fourfold pattern of risk attitudes for all participants. Panel (b) plots the same pattern separately for participants who pass the ADLE test versus those who fail. In both panels, the x -axis is the payout probability p in the lotteries $(\$25, p; \$0)$ and $(-\$25, p; \$0)$, and the y -axis is the difference between the CE and the expected value, with positive values indicating risk seeking and negative values indicating risk aversion. Error bars correspond to ± 2 standard errors.

D Appendix: Additional results in student-sample experiment

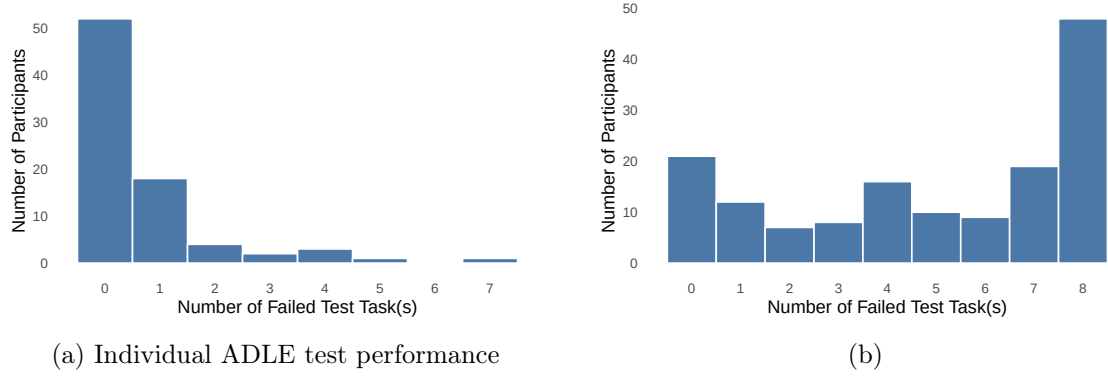


Figure D.1: Participants' performance on the ADLE test. Panel (a) is for the student sample. Panel (b) is for the Prolific sample.

Table D.1: Certainty equivalent deviations by the ADLE test performance.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
Panel A: Gain domain									
10	0	G10	52	4.154	3.5	0.000	0.000	0.111	0.139
10	1	G10	29	5.362	4.5	0.000	0.000		
25	0	G25	52	7.481	6.5	0.035	0.040	0.878	0.580
25	1	G25	29	7.362	6.5	0.189	0.334		
50	0	G50	52	11.288	12.5	0.010	0.002	0.214	0.241
50	1	G50	29	10.224	9.5	0.008	0.006		
75	0	G75	52	14.846	15.5	0.000	0.000	0.004	0.011
75	1	G75	29	11.741	10.5	0.000	0.000		
90	0	G90	52	19.058	20.5	0.000	0.000	0.013	0.025
90	1	G90	29	15.741	16.5	0.000	0.000		
Panel B: Loss domain									
10	0	L10	52	-4.769	-3.5	0.000	0.000	0.004	0.049
10	1	L10	29	-7.776	-5.5	0.000	0.000		
25	0	L25	52	-9.269	-8.0	0.000	0.000	0.508	0.453
25	1	L25	29	-9.879	-10.5	0.000	0.001		
50	0	L50	52	-12.769	-12.5	0.576	0.288	0.131	0.047
50	1	L50	29	-14.052	-15.5	0.042	0.021		
75	0	L75	52	-17.423	-18.5	0.074	0.229	0.253	0.257
75	1	L75	29	-16.155	-17.5	0.030	0.059		
90	0	L90	52	-21.077	-22.5	0.001	0.000	0.383	0.563
90	1	L90	29	-20.328	-21.5	0.018	0.007		

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. T_{fail} = indicator for ADLE-test failure (1 = fail, 0 = pass). All values rounded to three decimals.

Table D.2: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.619*** (0.024)	0.682*** (0.026)	0.611*** (0.026)	0.668*** (0.026)	0.706*** (0.023)	0.776*** (0.025)	0.710*** (0.024)	0.768*** (0.025)
F_T		0.019 (0.013)		0.024 (0.014)		0.047*** (0.013)		0.059*** (0.014)
$p \times F_T$		-0.091*** (0.018)		-0.112*** (0.019)		-0.102*** (0.017)		-0.113*** (0.018)
$F_{Compreh}$			-0.017 (0.033)	-0.038 (0.033)			-0.031 (0.031)	-0.082* (0.033)
$p \times F_{Compreh}$			0.038 (0.044)	0.135** (0.045)			-0.020 (0.042)	0.078 (0.043)
Intercept	0.127*** (0.018)	0.114*** (0.019)	0.130*** (0.019)	0.118*** (0.019)	0.178*** (0.017)	0.145*** (0.019)	0.184*** (0.018)	0.154*** (0.019)
Observations	405	405	405	405	405	405	405	405

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; F_T indicates the number of ADLE tasks failed by the participant; $F_{Compreh}$ indicates the number of comprehension questions failed by the participant. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random effects.

Table D.3: Mixed-effects Tobit regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.622*** (0.024)	0.703*** (0.029)	0.619*** (0.026)	0.695*** (0.029)	0.710*** (0.023)	0.772*** (0.028)	0.716*** (0.025)	0.771*** (0.029)
D_T		0.061 (0.036)		0.066 (0.037)		0.111** (0.036)		0.119** (0.036)
$p \times D_T$		-0.228*** (0.048)		-0.240*** (0.049)		-0.173*** (0.047)		-0.174*** (0.048)
$D_{Compreh}$			-0.018 (0.050)	-0.036 (0.050)			-0.027 (0.049)	-0.058 (0.049)
$p \times D_{Compreh}$			0.018 (0.067)	0.082 (0.067)			-0.039 (0.065)	0.006 (0.065)
Intercept	0.124*** (0.018)	0.103*** (0.022)	0.127*** (0.019)	0.106*** (0.022)	0.175*** (0.017)	0.135*** (0.022)	0.179*** (0.019)	0.141*** (0.022)
Observations	405	405	405	405	405	405	405	405

Notes: The dependent variable is $|CE|/25$, the normalized absolute certainty equivalent. Reported coefficients are mixed-effects Tobit models with double censoring at 0.02 and 1.02. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random intercepts.

E Appendix: Additional results in Experiment Binary choices

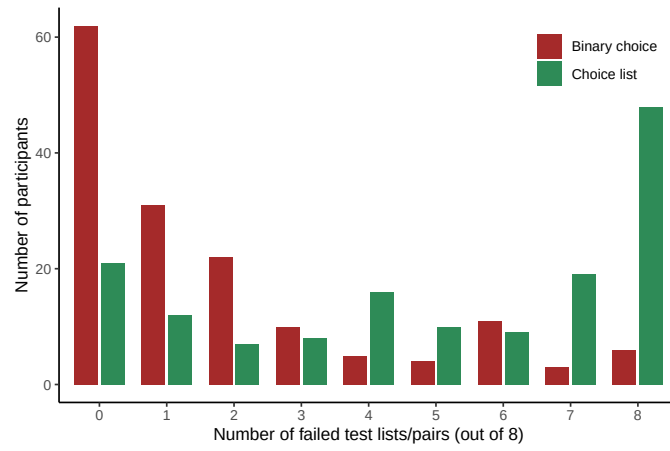


Figure E.1: Test performance comparison between the main price-list experiment and the binary-choice experiment. *Notes:* We say that a participant passes a test choice pair if she passes both binary choices (e.g., choosing $(\$2.90, 0.10; \$2.10)$ over $\$2$ and choosing $\$3$ over $(\$2.90, 0.10; \$2.10)$).

F Appendix: Pooling online experiments

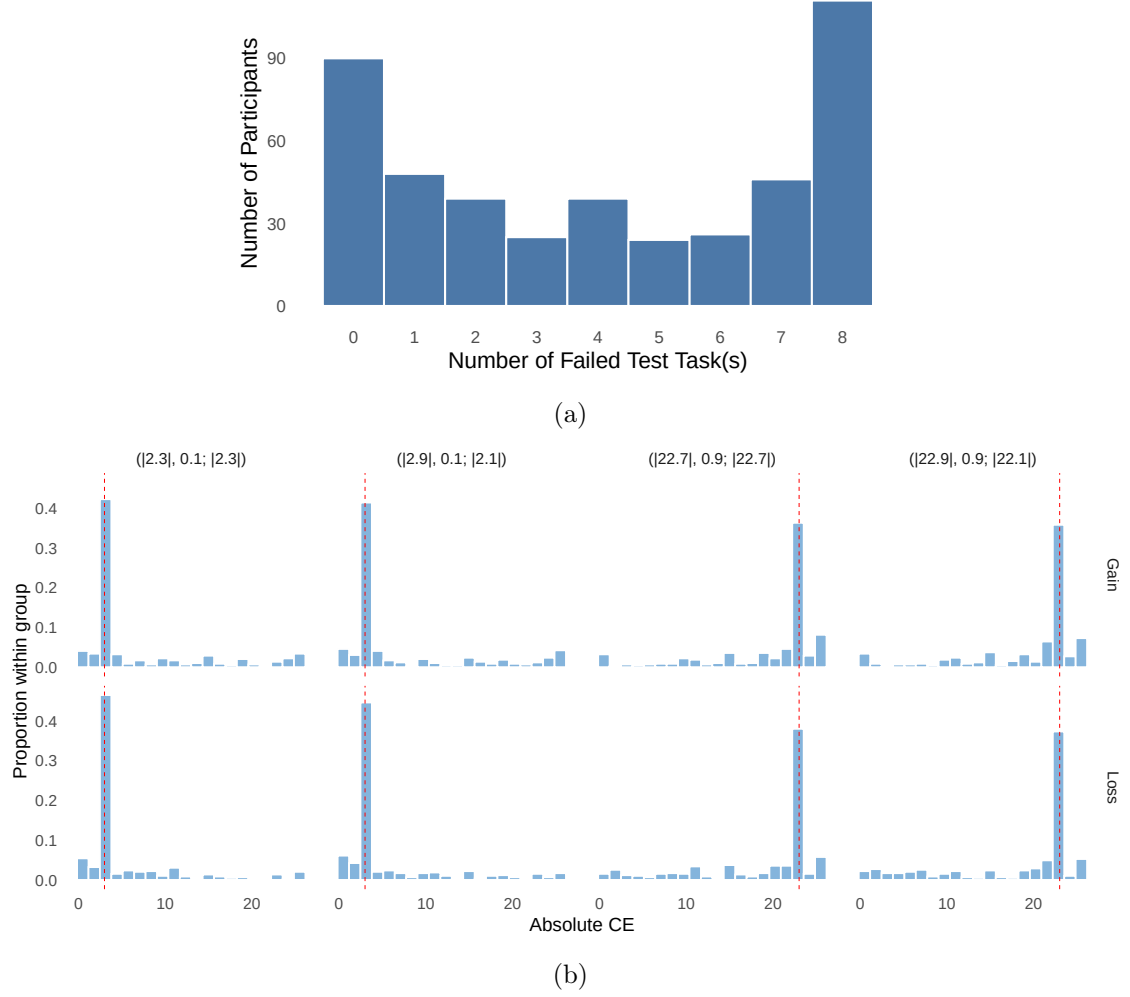


Figure F.1: Performance on the ADLE test. *Notes:* Panel (a) shows the distribution of the number of failed ADLE tasks by each participant. The sample consists of 448 participants. Panel (b) plots the distribution of individual CEs—i.e., switching positions—across the eight test lotteries, separately for gain and loss domains. Red dashed lines indicate the theoretically correct switching position for each list.

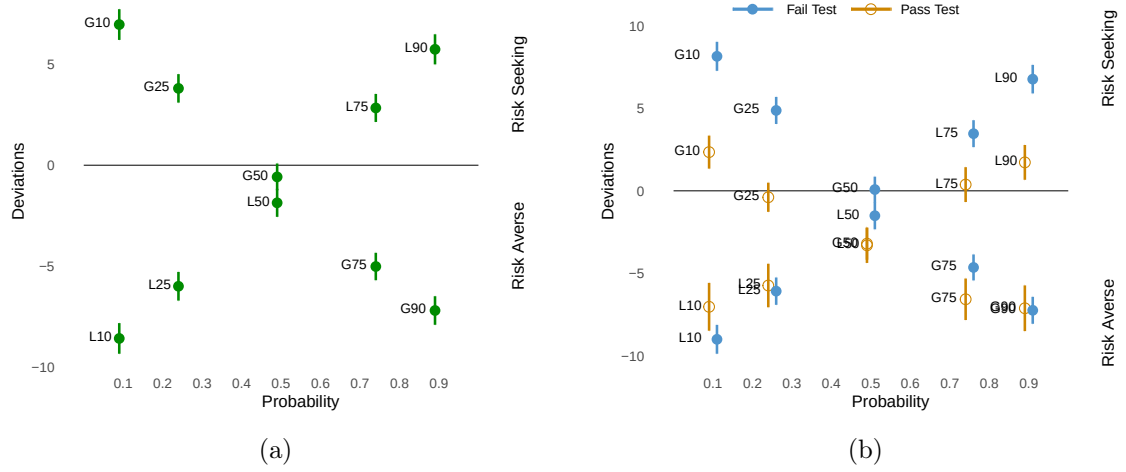


Figure F.2: The fourfold pattern of risk attitudes and ADLE test performance. *Notes:* Panel (a) plots the fourfold pattern of risk attitudes for all participants. Panel (b) plots the same pattern separately for participants who pass and who fail the ADLE test. In both panels, the x -axis is the payout probability p in the lotteries $(\$25, p; \$0)$ and $(-\$25, p; \$0)$, and the y -axis is the deviation of the CE from the expected value, with positive values indicating risk seeking and negative values indicating risk aversion. Error bars correspond to ± 2 standard errors. Table F.1 reports the corresponding summary statistics.

Table F.1: Deviations of certainty equivalents from expected values by the ADLE test performance.

P	T _{fail}	Label	N	Mean	Median	T1	W1	T2	W2
<i>Panel A: Gain domain</i>									
10	0	G10	90	4.844	3.5	0.000	0.000		
10	1	G10	358	10.645	9.5	0.000	0.000	0.000	0.000
25	0	G25	90	6.111	4.5	0.381	0.689		
25	1	G25	358	11.369	9.5	0.000	0.000	0.000	0.000
50	0	G50	90	9.300	9.5	0.000	0.000		
50	1	G50	358	12.578	12.5	0.842	0.932	0.000	0.000
75	0	G75	90	11.933	12.0	0.000	0.000		
75	1	G75	358	13.860	14.5	0.000	0.000	0.023	0.018
90	0	G90	90	15.389	15.5	0.000	0.000		
90	1	G90	358	15.266	15.5	0.000	0.000	0.890	0.721
<i>Panel B: Loss domain</i>									
10	0	L10	90	-9.522	-9.5	0.000	0.000		
10	1	L10	358	-11.492	-10.5	0.000	0.000	0.038	0.065
25	0	L25	90	-12.233	-10.5	0.000	0.000		
25	1	L25	358	-12.575	-10.5	0.000	0.000	0.701	0.823
50	0	L50	90	-15.800	-15.5	0.000	0.000		
50	1	L50	358	-14.006	-14.5	0.000	0.000	0.039	0.057
75	0	L75	90	-18.122	-19.5	0.476	0.908		
75	1	L75	358	-15.039	-15.5	0.000	0.000	0.000	0.002
90	0	L90	90	-20.778	-22.5	0.001	0.014		
90	1	L90	358	-15.737	-16.5	0.000	0.000	0.000	0.000

Notes: T1 = one-sample t-test vs. prediction. W1 = Wilcoxon signed-rank test vs. prediction. T2 = two-sample t-test across test-performance groups. W2 = Wilcoxon rank-sum test across test-performance groups. All values rounded to three decimals. See corresponding Figure F.2b.

Table F.2: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.280*** (0.016)	0.510*** (0.035)	0.359*** (0.023)	0.529*** (0.036)	0.274*** (0.016)	0.537*** (0.035)	0.337*** (0.023)	0.546*** (0.036)
D_T		0.273*** (0.032)		0.224*** (0.033)		0.104** (0.033)		0.117*** (0.035)
$p \times D_T$		-0.288*** (0.039)		-0.255*** (0.041)		-0.329*** (0.040)		-0.313*** (0.042)
$D_{Compreh}$			0.184*** (0.026)	0.129*** (0.027)			-0.005 (0.027)	-0.034 (0.028)
$p \times D_{Compreh}$			-0.149*** (0.032)	-0.087** (0.033)			-0.119*** (0.032)	-0.042 (0.033)
Intercept	0.344*** (0.013)	0.126*** (0.029)	0.247*** (0.019)	0.097*** (0.029)	0.426*** (0.013)	0.343*** (0.030)	0.429*** (0.019)	0.351*** (0.030)
Observations	2240	2240	2240	2240	2240	2240	2240	2240

Notes: Dependent variable = $|CE|/25$ (normalized absolute certainty equivalent). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Standard errors in parentheses. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

Table F.3: Mixed-effects regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.280*** (0.016)	0.506*** (0.025)	0.349*** (0.020)	0.511*** (0.026)	0.274*** (0.016)	0.523*** (0.026)	0.341*** (0.020)	0.526*** (0.026)
F_T		0.044*** (0.004)		0.038*** (0.004)		0.010* (0.004)		0.011* (0.005)
$p \times F_T$		-0.055*** (0.005)		-0.051*** (0.005)		-0.061*** (0.005)		-0.059*** (0.005)
$F_{Compreh}$			0.066*** (0.008)	0.036*** (0.008)			0.004 (0.008)	-0.004 (0.009)
$p \times F_{Compreh}$			-0.052*** (0.009)	-0.016 (0.010)			-0.054*** (0.010)	-0.008 (0.010)
Intercept	0.344*** (0.013)	0.155*** (0.021)	0.263*** (0.016)	0.144*** (0.021)	0.426*** (0.013)	0.386*** (0.021)	0.421*** (0.017)	0.387*** (0.022)
Observations	2240	2240	2240	2240	2240	2240	2240	2240

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE_{ij}|/25$, for lottery j of participant i . p is the lottery's payout probability; F_T indicates the number of ADLE tasks participants failed; $F_{Compreh}$ indicates the number of comprehension questions participants failed. Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The first four columns (1–4) report results for the gain domain; the last four columns (5–8) report results for the loss domain. All models use random effects with participant-level clustering.

Table F.4: Mixed-effects Tobit regressions on normalized absolute certainty equivalents

	<i>Gain Domain</i>				<i>Loss Domain</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
p	0.304*** (0.018)	0.535*** (0.039)	0.391*** (0.026)	0.557*** (0.040)	0.300*** (0.018)	0.557*** (0.040)	0.366*** (0.026)	0.568*** (0.040)
D_T		0.282*** (0.037)		0.226*** (0.039)		0.110** (0.038)		0.123** (0.039)
$p \times D_T$		-0.291*** (0.044)		-0.252*** (0.046)		-0.324*** (0.044)		-0.305*** (0.047)
$D_{Compreh}$			0.205*** (0.030)	0.148*** (0.031)			-0.005 (0.031)	-0.036 (0.032)
$p \times D_{Compreh}$			-0.165*** (0.036)	-0.101** (0.037)			-0.127*** (0.036)	-0.051 (0.038)
Intercept	0.327*** (0.015)	0.102** (0.033)	0.219*** (0.022)	0.069* (0.033)	0.418*** (0.015)	0.331*** (0.034)	0.421*** (0.022)	0.340*** (0.034)
Observations	2240	2240	2240	2240	2240	2240	2240	2240

Notes: The dependent variable is the normalized absolute certainty equivalent, $|CE|/25$. Reported coefficients are from mixed-effects Tobit models with double censoring at 0.02 and 1.02. Standard errors are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $D_T = 1$ indicates participants who failed the ADLE test; $D_{Compreh} = 1$ indicates participants who failed the comprehension questions. Columns (1)–(4) report results for the gain domain, and Columns (5)–(8) report results for the loss domain. All models include participant-level random intercepts.

Table F.5: The proportion of participants choosing the lottery under different conditions.

	(\$25,0.1;0) vs \$3		(\$25,0.9;0) vs \$22		(-\$25,0.1;0) vs -\$3		(-\$25,0.9;0) vs -\$22	
	PL	BC	PL	BC	PL	BC	PL	BC
Pass (n=90)	0.522	0.344	0.244	0.189	0.267	0.267	0.478	0.422
Fail (n=358)	0.774	0.416	0.268	0.223	0.279	0.190	0.542	0.673
P-value	0.000	0.231	0.690	0.567	0.895	0.111	0.290	0.000

Notes: Pass/fail indicates performance on the ADLE test; PL = price list choice, BC = binary choice. Sample sizes (n) for Pass/Fail groups are shown in parentheses. P-values from Fisher's Exact Test compare lottery choice proportions between subgroups (values rounded to 3 decimal places).

Table F.6: Proportion of consistent choices (retest vs. original) by ADLE test performance.

	Consistent Choice Proportion			
	(\$25,0.1;0)	(\$25,0.9;0)	(-\$25,0.1;0)	(-\$25,0.9;0)
Pass	21/51	16/39	18/41	28/49
Fail	69/181	61/177	59/180	67/178
p -value	0.746	0.464	0.205	0.021

Notes: Values are formatted as “count of consistent choices / valid sample size”. Pass/fail refers to performance on the ADLE test. p -values from Fisher's Exact Test compare consistency proportions between the two subgroups.

Table F.7: OLS regressions on Failure frequency in the ADLE test

	<i>ADLE tasks failure frequency</i>		
	(1)	(2)	(3)
$F_{\text{Compreh.}}$	0.742*** (0.084)	0.709*** (0.082)	0.702*** (0.083)
Complexity	0.292* (0.123)	0.299* (0.121)	0.306* (0.121)
CRT	0.292* (0.130)	0.262* (0.127)	0.254* (0.128)
PL-Rep		−0.812*** (0.170)	−0.818*** (0.171)
PL&BC		0.012 (0.125)	0.015 (0.125)
Female			−0.201 (0.262)
Age			0.008 (0.010)
Intercept	1.847*** (0.450)	2.519*** (0.568)	2.334** (0.706)
Observations	448	448	448

Notes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. $F_{\text{Compreh.}}$ denotes the number of failed comprehension questions. Complexity refers to participants' self-reported perception of the overall difficulty of the price lists (range: 1–6). CRT indicates correct answers in the Cognitive Reflection Test (range: 0–3). PL-Rep denotes the number of times that choices between the initial and the repeated price list were consistent (range: 0–4). PL&BC represents the number of times that choices implied in the price lists were consistent with their corresponding binary choices (range: 0–4). Standard errors are in parentheses.

G Appendix: Experimental Design

G.1 Screenshots of the Main Experimental Interface

Consent

Welcome to our study. We estimate that it will take approximately 25 minutes to complete.

At the end of the experiment, one of your decisions from the main task will be randomly selected to determine your bonus. Depending on your choices, your bonus will range between **€5 and €30**.

When you clicked the link to join, a computer program randomly determined which decision would determine your bonus payment. Your choices cannot influence this outcome—it has already been set.

This study does not involve any deception. All information provided is truthful, and nothing is being withheld from you. Everything will be conducted exactly as described in the instructions.

Please give your full attention throughout the study. Refrain from using your phone, browsing the internet, or engaging in any other activities.

By clicking the button below, you give your informed consent to participating in this study. Your participation is voluntary, and you may leave the study at any time. **Please note: Payment will only be provided if you complete the study.**

I consent to the statements above and wish to participate this study.

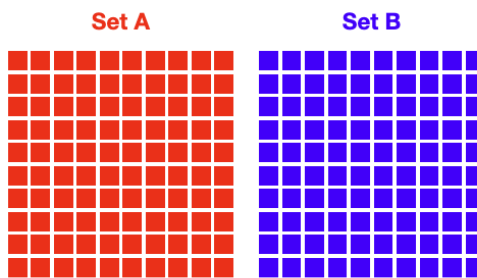
I do not consent to the statements above and want to leave this study.

NEXT

General Instructions

Boxes With Money

- In each of several tasks, we will give you an INITIAL sum of money.
- You will then choose which set of BOXES – Set A or Set B – the computer should open.



- Each box contains either a POSITIVE or NEGATIVE amount of money (or nothing). When the computer opens one box from your chosen set, the amount of money in the **opened** box will be added (or subtracted) from your INITIAL money to determine your FINAL EARNINGS.

[Continue Instructions](#)

General Instructions

The Decision Table

- The two sets of boxes will be shown in a TABLE as illustrated below. For each set, the number of boxes (e.g., 100, 75, or 25 boxes) is listed at the top, and the corresponding monetary value (e.g., \$7, \$20, or \$0) for that count is displayed in the table row.

Set A	Set B	
100 Boxes	75 Boxes	25 Boxes
\$7.00	\$20.00	\$0.00

- In the example above, Set A consists of 100 boxes, each containing \$7; Set B consists of 75 boxes with \$20 in each box and 25 boxes with \$0 in each box.
- In the example below, Set A consists of 100 boxes, all containing -\$3 (negative \$3). Set B consists of 75 boxes with -\$12 (negative \$12) in each box and 25 boxes with \$0 in each box.

Set A	Set B	
100 Boxes	75 Boxes	25 Boxes
-\$3.00	-\$12.00	\$0.00

- Your task is to click on the table to choose which set of boxes (A or B) you want the computer to use to determine your payment. Clicking on the table will highlight one set in yellow. The set highlighted in yellow will be used to determine your FINAL EARNINGS.

Set A	Set B	
100 Boxes	75 Boxes	25 Boxes
\$7.00	\$20.00	\$0.00

- In the example above, you have highlighted Set A, and your payment will be based on that set.

Instruction – A Random Box

- In the upcoming tasks, the computer will RANDOMLY select one of the 100 boxes from the set you choose. Every box in your chosen set is EQUALLY likely to be selected. If the selected box contains a positive amount, that amount will be ADDED to your initial money; if it is negative, the amount will be SUBTRACTED from your initial money.
- Example: In the example below, each set contains 100 boxes. For Set A, every one of the 100 boxes contains \$4.00. Therefore, if you choose Set A, you have a 100% chance of having \$4.00 added to your initial money. In Set B, 50 boxes contain \$16.00 and the remaining 50 contain \$0.00. Thus, if you choose Set B, there is a 50% chance that \$16.00 will be added and a 50% chance that \$0.00 will be added.

Set A	Set B	
100 Boxes	50 Boxes	50 Boxes
\$4.00	\$16.00	\$0.00

- Example: In the example below, each set also contains 100 boxes. For Set A, all 100 boxes contain -\$6.00. If you choose Set A, you have a 100% chance of having \$6.00 subtracted from your initial money. In Set B, 50 boxes contain -\$8.00 and the other 50 contain \$0.00. Therefore, if you choose Set B, there is a 50% chance that \$8.00 will be subtracted and a 50% chance that \$0.00 will be subtracted.

Set A	Set B	
100 Boxes	50 Boxes	50 Boxes
-\$6.00	-\$8.00	\$0.00

Continue Instructions

Here are some examples to help you answer the comprehension questions in next page correctly.

Suppose in the following two sets, you chose Set B.

Set A	Set B	
100 Boxes	60 Boxes	40 Boxes
\$2.00	\$5.00	\$0.00

Q1: What is the chance that \$5 is added to your earnings?

- ☐ 0 in 100 (0%)
- ☐ 40 in 100 (40%)
- ☒ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

The correct answer is C for this question. This is because you selected Set B. In Set B, the chance that \$5 is added to your earnings is 60 in 100 (60%).

Q2: What is the chance that \$2 is added to your earnings?

- ☒ 0 in 100 (0%)
- ☐ 40 in 100 (40%)
- ☐ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

The correct answer is A for this question. This is because you selected Set B, not Set A. Thus, the chance that \$2 (Set A) is added to your earnings is 0 in 100 (0%).

Q3: What is the chance that \$0 is added to your earnings?

- ☐ 0 in 100 (0%)
- ☒ 40 in 100 (40%)
- ☐ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

The correct answer is B for this question. This is because you selected Set B. In Set B, the chance that \$0 is added to your earnings is 40 in 100 (40%).

By clicking "Continue," you will begin the real comprehension questions. Please note that you can earn an extra **reward of \$0.15** if you answer a comprehension question correctly on your first try.

Continue

Comprehension Questions - A Random Box

Suppose in the following two sets, you chose Set B. Please answer the three questions below:

Set A	Set B	
100 Boxes	40 Boxes	60 Boxes
\$5.00	\$10.00	\$0.00

Q1: What is the chance that \$10 is added to your earnings?

- ☐ 0 in 100 (0%)
- ☒ 40 in 100 (40%)
- ☐ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

Correct!

Q2: What is the chance that \$4 is added to your earnings?

- ☒ 0 in 100 (0%)
- ☐ 40 in 100 (40%)
- ☐ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

Correct!

Q3: What is the chance that \$5 is added to your earnings?

- ☒ 0 in 100 (0%)
- ☐ 40 in 100 (40%)
- ☐ 60 in 100 (60%)
- ☐ 100 in 100 (100%)

Submit Quiz

General Instructions

Decision Screen: Choosing a Set of Boxes

- In the actual experiment, you will face a sequence of questions to choose between Set A and Set B. Each question is presented as a different row in the table.
- **Example:** In the first row (Question 1) in the example below, Set A has 100 boxes containing \$10 while Set B has 40 boxes containing \$10 and 60 boxes containing \$0. However, in the second row (Question 2), Set A has 100 boxes containing \$9, while Set B has 40 boxes containing \$10 and 60 boxes containing \$0. Additional rows display other Question of Set A and Set B.

Question	Set A	Set B	
	100 Boxes	40 Boxes	60 Boxes
1	\$10.00	\$10.00	\$0.00
2	\$9.00	\$10.00	\$0.00
3	\$8.00	\$10.00	\$0.00
4	\$7.00	\$10.00	\$0.00
5	\$6.00	\$10.00	\$0.00
6	\$5.00	\$10.00	\$0.00
7	\$4.00	\$10.00	\$0.00
8	\$3.00	\$10.00	\$0.00
9	\$2.00	\$10.00	\$0.00
10	\$1.00	\$10.00	\$0.00

SUBMIT

- You will make a choice between Set A and Set B for all questions by clicking on the table and highlighting your preferred set in each row. Please try clicking the table above.
- When making your choices, the computer will limit you to switching from Set A to Set B only once across the table (although you may choose exclusively Set A or exclusively Set B for all rows). You may click on the table as many times as needed until you are satisfied with your selections. Then, press the **SUBMIT** button to finalize your decision.
- At the end of the experiment, the computer will randomly select one row from the table (each row is equally likely) and pay you based on your choice in that row. This means you should carefully consider your decision for every row (every question), as any row could determine your final payment.

Continue Instructions

Here are some examples to help you answer the comprehension questions in next page correctly.

Suppose you made the decisions shown in the table below.

Question	Set A	Set B	
	100 Boxes	60 Boxes	40 Boxes
1	\$5.00	\$5.00	\$0.00
2	\$4.00	\$5.00	\$0.00
3	\$3.00	\$5.00	\$0.00
4	\$2.00	\$5.00	\$0.00
5	\$1.00	\$5.00	\$0.00

Q1: If Question 2 of this table is selected to determine your payment, which statement is correct?

- ☐ You will receive \$5.
- ☒ You will receive \$4.
- ☐ You will receive \$3.
- ☐ You will receive \$0.

The correct answer is B. This is because you chose Set A for Question 2, and Set A adds \$4 to your earnings for sure (100%).

Q2: If Question 4 of this table is selected to determine your payment, which statement is correct?

- ☐ You will receive \$4.
- ☐ You will receive \$2.
- ☒ You will have 60% chance of receiving \$5 and 40% chance of receiving \$0.
- ☐ You will have 50% chance of receiving \$5 and 50% chance of receiving \$0.

The correct answer is C. This is because you chose Set B for Question 4. Its Set B could add \$5 to your earnings with a chance of 60 in 100 (60%), and add \$0 with a chance of 40 in 100 (40%).

Q3: Which statement best describes your choices?

- ☐ You chose Set A over Set B for Question 5.
- ☐ You chose Set A over Set B for Question 4.
- ☒ You chose Set A over Set B for Question 3.
- ☐ You chose Set B over Set A for Question 2.

The correct answer is C. This is because you chose Set A for Questions 1 to 3, and then chose Set B for remaining Questions 4 and 5.

By clicking "Continue," you will begin the real comprehension questions. Please note that you can earn an extra **reward of \$0.15** if you answer a comprehension question correctly on your first try.

Continue

Comprehension Questions – Decision Table

Suppose you made the decisions shown in the table below. Please answer the three questions below:

	Set A	Set B	
Question	100 Boxes	40 Boxes	60 Boxes
1	\$10.00	\$10.00	\$0.00
2	\$9.00	\$10.00	\$0.00
3	\$8.00	\$10.00	\$0.00
4	\$7.00	\$10.00	\$0.00
5	\$6.00	\$10.00	\$0.00
6	\$5.00	\$10.00	\$0.00
7	\$4.00	\$10.00	\$0.00
8	\$3.00	\$10.00	\$0.00
9	\$2.00	\$10.00	\$0.00
10	\$1.00	\$10.00	\$0.00

Q1: If Question 3 of this table is selected to determine your payment, which statement is correct?

- ☐ You will receive \$4.
- ☐ You will receive \$10.
- ☒ You will receive \$8.
- ☐ You will receive \$0.

Correct!

Q2: If Question 6 of this table is selected to determine your payment, which statement is correct?

- ☐ You will receive \$4.
- ☐ You will receive \$5.
- ☐ You will have 50% chance of receiving \$10 and 50% chance of receiving \$0.
- ☒ You will have 40% chance of receiving \$10 and 60% chance of receiving \$0.

Correct!

Q3: Which statement best describes your choices?

- ☐ You chose Set A over Set B for Question 9.
- ☐ You chose Set A over Set B for Question 5.
- ☐ You chose Set A over Set B for Question 4.
- ☐ You chose Set B over Set A for Question 2.

Submit Quiz

General Instructions

Multiple Tables

- Throughout the experiment, you will be presented with several tables. Each table displays a different initial amount of money along with different monetary amounts in the boxes arranged in rows. You must make a decision for every question (row) of each table.
- You will receive a BONUS based on your choices. Specifically, the computer will randomly choose one table and then randomly select one question (row) from that table to determine your bonus payment.
- Since you will not know which decision will be used to calculate your payment, please treat each choice as if it alone determines your final payment.

[Continue Instructions](#)

Choice task screenshot. In this example, a participant switch from Option A to Option B between Row 17 and Row 18.

Initial Money: \$5.00

- Please select which Set (A or B) you'd prefer for each row of the table (each question of the problem) and click the Submit button.
- If this task is selected for payment, the computer will randomly select one row (one question) and use your choice in this row to determine your earnings.
- You will be paid \$5 plus the value of one of the boxes from the Set you selected, randomly chosen by the computer.

	Set A	Set B	
Question	100 Boxes	50 Boxes	50 Boxes
1	\$25.00	\$25.00	\$0.00
2	\$24.00	\$25.00	\$0.00
3	\$23.00	\$25.00	\$0.00
4	\$22.00	\$25.00	\$0.00
5	\$21.00	\$25.00	\$0.00
6	\$20.00	\$25.00	\$0.00
7	\$19.00	\$25.00	\$0.00
8	\$18.00	\$25.00	\$0.00
9	\$17.00	\$25.00	\$0.00
10	\$16.00	\$25.00	\$0.00
11	\$15.00	\$25.00	\$0.00
12	\$14.00	\$25.00	\$0.00
13	\$13.00	\$25.00	\$0.00
14	\$12.00	\$25.00	\$0.00
15	\$11.00	\$25.00	\$0.00
16	\$10.00	\$25.00	\$0.00
17	\$9.00	\$25.00	\$0.00
18	\$8.00	\$25.00	\$0.00
19	\$7.00	\$25.00	\$0.00
20	\$6.00	\$25.00	\$0.00
21	\$5.00	\$25.00	\$0.00
22	\$4.00	\$25.00	\$0.00
23	\$3.00	\$25.00	\$0.00
24	\$2.00	\$25.00	\$0.00
25	\$1.00	\$25.00	\$0.00

You have finished the first part of the experiment.

How would you rate the complexity of the choice task you completed?

Very easy to understand

Esay to understand

Somewhat easy to understand

Somewhat difficult to understand

Difficult to understand

Very difficult to understand

NEXT

Binary Choice Example

Imagine you have been given \$30. Which of the following would you prefer?

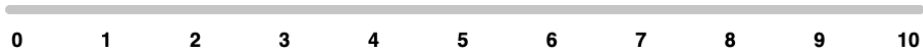
Lose \$25 with probability 0.1; 0 with probability 0.9

Lose \$3 with certainty

NEXT

How do you see yourself: are you generally a person who is fully prepared to take risks or do you try to avoid taking risks?

Please drag the slider, where the **value 0** means: 'not at all willing to take risks' and the **value 10** means: 'very willing to take risks'.



If John can drink one barrel of water in 6 days, and Mary can drink one barrel of water in 12 days, how long would it take them to drink one barrel of water together?

Answer: day(s).

A man buys a pig for \$60, sells it for \$70, buys it back for \$80, and sells it finally for \$90. How much has he made?

Answer: dollar(s).

Simon decided to invest \$8,000 in the stock market one day early in 2008. Six months after he invested, on July 17, the stocks he had purchased were down 50%. Fortunately for Simon, from July 17 to October 17, the stocks he had purchased went up 75%.

At this point, Simon has:

Answer: dollar(s).

This concludes the MAIN portion of the experiment.
Thank you for your participation!

To finish the study, please answer the following questions about yourself.

Q1: Please indicate your gender

Male

Female

Q2: Please drag the slider to indicate your age

You are "Use the slider!" years old.



Q3: What level of education do you complete?

No formal education

Elementary school

Middle school

High school

Undergraduate school

Postgraduate school

Q4: Do you have any comments, suggestions, or feedback about your participation, or anything else you'd like to share about this experiment? (Optional)

NEXT

G.2 Screenshots of the Mirror Task

Initial Money: \$30.00

- Please select which Set (A or B) you'd prefer for each row of the table (each version of the problem) and click the Submit button.
- If this task is selected for payment, the computer will randomly select one row (one version) and use your choice in this row to determine your earnings.
- You will be paid \$30 minus the value of all of the boxes from the Set you selected, added up and divided by 100.

	Set A	Set B	
Question	100 Boxes	50 Boxes	50 Boxes
1	-\$1.00	-\$25.00	\$0.00
2	-\$2.00	-\$25.00	\$0.00
3	-\$3.00	-\$25.00	\$0.00
4	-\$4.00	-\$25.00	\$0.00
5	-\$5.00	-\$25.00	\$0.00
6	-\$6.00	-\$25.00	\$0.00
7	-\$7.00	-\$25.00	\$0.00
8	-\$8.00	-\$25.00	\$0.00
9	-\$9.00	-\$25.00	\$0.00
10	-\$10.00	-\$25.00	\$0.00

Figure G.1: (Partial) Screenshot of the deterministic mirror task. *Notes:* This is a loss-domain mirror list from the mirror task, where the participant switches from the sure-amount option to the mirror option between Question 7 and Question 8.

G.3 Screenshots of the Cognitive Uncertainty Question

13	\$13.00	\$25.00	\$0.00
14	\$12.00	\$25.00	\$0.00
15	\$11.00	\$25.00	\$0.00
16	\$10.00	\$25.00	\$0.00
17	\$9.00	\$25.00	\$0.00
18	\$8.00	\$25.00	\$0.00
19	\$7.00	\$25.00	\$0.00
20	\$6.00	\$25.00	\$0.00
21	\$5.00	\$25.00	\$0.00
22	\$4.00	\$25.00	\$0.00
23	\$3.00	\$25.00	\$0.00
24	\$2.00	\$25.00	\$0.00
25	\$1.00	\$25.00	\$0.00

You chose **Set A** over **Set B** in Versions 1 to 17 and **Set B** over **Set A** in Versions 18 to 25.
How certain do you feel about your decision?

|

|

●

very uncertain **completely certain**

I am **60%** certain about my choices in the Table.

Figure G.2: Screenshot of the cognitive uncertainty task.

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G.4 Screenshots of the Binary-Choice Question

Initial Money: \$5.00

- Please select which Set (A or B) you'd prefer and click the Submit button.
- If this task is selected for payment, the computer will use your choice to determine your earnings.
- You will be paid \$5 plus the value of one of the boxes from the Set you selected, randomly chosen by the computer.

Gain Question 1

Set A	Set B	
100 Boxes	90 Boxes	10 Boxes
\$22.00	\$25.00	\$0.00

SUBMIT

Figure G.3: Screenshot of the binary-choice task.